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MASTER'S DEGREE IN BIOINFORMATICS

**A One Health Decision Support System for the
assessment and prediction of food contamination
risk**

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Abstract

Motivation: Accredited food testing laboratories generate extensive volumes of analytical data as part of routine compliance monitoring mandated by Regulation (EU) 2023/915. In practice, this data is used mainly for individual pass or fail assessments, leaving a valuable epidemiological resource largely untapped. Risk prioritization still relies on individual expert judgment rather than on a structured, data-driven process, and what is missing is a tool able to turn raw laboratory records into a form that supports systematic and reproducible decision-making. A further gap concerns the translation of compliance data into biological relevance. Existing regulatory frameworks assess contaminants individually, so they disregard shared molecular targets and the combinatorial toxicity that can arise from simultaneous dietary co-exposure, and no operator-facing tool currently links laboratory findings to a One Health perspective spanning human, environmental, and food safety domains. Closing this gap requires statistical and machine-learning modelling of the laboratory record alongside gene-pathway network analysis of the contaminants it contains.

Results: A Decision Support System (DSS) covering the full analytical workflow was validated on 182,033 certified results from Neutron Laboratories (Modena, Italy) across six regulatory contaminant sections. A CatBoost classifier achieved a test PR-AUC of 0.811 and a ROC-AUC of 0.950, a 4.4x lift over baseline. SHAP analysis showed that the structural breadth of non-conformity across the submatrix and contaminant portfolios, rather than the direct history of the specific pair being scored, formed the two dominant predictors. Of the 3,495 modelled pairs, 127 were classified HIGH risk, 248 MEDIUM, 1,143 LOW, and 1,977 INSUFFICIENT. Bidirectional blind validation confirmed model stability in both temporal directions, prospectively on January to May 2026 (PR-AUC 0.836, ROC-AUC 0.994) and retrospectively on the whole of 2023 (PR-AUC 0.862, ROC-AUC 0.988), with both windows exceeding the training reference metrics under frozen parameters. A One Health analysis then linked the 47 highest-priority contaminants to 405 gene targets across 263 pathways via the Comparative Toxicogenomics Database and KEGG enrichment. *A2M*, *ABCB1* and *AARS1* emerged as the top shared hub genes, connecting lipid metabolism and atherosclerosis, cancer-related pathways, and ABC transporters, and pointing to convergent toxicological mechanisms that remain invisible to single-contaminant regulatory assessment.

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Chapter 1

Introduction

1.1 Food safety as a global public health priority

Food safety constitutes a strategically relevant challenge in modern public health, sitting at the intersection of human medicine, environmental science, agricultural policy, industrial processing, and regulatory governance. Chemical contaminants pose a particular concern because they lack organoleptic indicators, and they can accumulate silently through long-term dietary exposure. The adverse health effects that follow are varied. Carcinogenicity, nephrotoxicity, and developmental neurotoxicity are among them, alongside endocrine disruption, reproductive toxicity, and immunotoxicity. These consequences often surface only years or even decades after exposure. For this reason, establishing direct causal relationships is inherently difficult. Food safety management, in turn, increasingly relies on precautionary principles and population-based risk assessment approaches.

The World Health Organization estimates that unsafe food causes approximately 600 million cases of foodborne illness and 420,000 deaths each year worldwide, with children under five bearing a disproportionate share despite their small fraction of the global population (World Health Organization, 2015). A WHO-commissioned data synthesis estimated over one million disability-adjusted life-years lost in 2010 to four foodborne chemical toxins alone, aflatoxin, dioxin, cyanide, and allergen exposure among them (Gibb et al., 2015). Even in Europe, contamination episodes still occur, as mycotoxins, heavy metals, and process-generated contaminants are among the recurring culprits, alongside persistent organic pollutants. This persists despite the region having one of the most comprehensive food safety regulatory systems in the world.

Chemical contamination behaves differently from microbiological hazards. Where the latter tend to produce acute, readily identifiable clinical symptoms, chemical exposure

is generally chronic, involving low concentrations accumulated over extended periods. The result is a genuine analytical and regulatory challenge. Contamination patterns emerge from a tangle of environmental factors, agricultural practices, and industrial processing methods, further complicated by storage conditions, climatic variability, and global supply chain dynamics.

Agricultural trade has grown steadily more international, and a single food product may now contain ingredients sourced from numerous countries, each with its own environmental conditions, soil composition, and farming practices, not to mention its own regulatory framework. Contamination profiles are becoming harder to predict using traditional regulatory approaches. What this points to is a clear need for surveillance systems that are data-driven and operate in real time.

Major contamination incidents, from the 2008 melamine adulteration crisis onward, have generated economic losses amounting to hundreds of millions of euros and undermined consumer confidence across entire food sectors, prompting regulatory responses across multiple jurisdictions (Gossner et al., 2009). Delayed detection compounds the damage further still, since it drives unnecessary food waste and reinforces the link between food safety, sustainability, and planetary health.

Contemporary food safety surveillance now relies heavily on integrated approaches, ones that can both analytically detect contaminants and interpret large, heterogeneous datasets (Wang et al., 2022; Soroushianfar et al., 2025). These approaches help identify emerging risks. They also make monitoring resources easier to allocate efficiently, and they supply the evidence base needed for informed decision making throughout the food supply chain.

1.2 Accredited laboratories as epidemiological data resources

Accredited analytical laboratories occupy a central, strategic place within the food safety framework. These institutions do more than assess regulatory compliance, as they consistently produce detailed occurrence data, and this data reveals how contaminants are distributed across food matrices, production processes, geographical origins, and temporal periods. Once such laboratory management system datasets are systematically collected and analyzed, they become valuable epidemiological resources. They can support contamination surveillance, exposure assessment, trend analysis over time, predictive modeling, and prioritization of operational risks (van den Bulk et al., 2025).

Industrial-scale laboratories often manage thousands of analytical determinations

daily. The resulting data, in granularity and volume, frequently surpasses what public monitoring programs obtain. Yet in typical operational environments, analytical results serve mainly one purpose, which is determining individual compliance decisions. Results are compared to defined maximum levels, assigned a conformity status, and reported to clients. The extensive statistical information embedded in historical records, contamination probability gradients, co-occurrence patterns, and the like, therefore tends to remain unexplored. Data used for regulatory compliance could equally serve to design risk-based sampling plans, establish early warning systems, and optimize monitoring plans, provided that appropriate analytical frameworks are put in place.

Neutron Laboratories, the accredited entity examined in this thesis, routinely conducts analytical activities spanning all major contaminant categories regulated under Regulation (EU) 2023/915, processing over 1,200 samples daily. This study's primary conceptual aim is to turn these historical records from mere compliance archives into actionable intelligence for risk management.

1.3 The European regulatory framework for food safety

The European Union built a food safety governance system over time, shaped in direct response to crises including Bovine Spongiform Encephalopathy in the 1990s, the 1999 Belgian dioxin contamination, and the Sudan Red adulteration affair. Regulation (EC) No 178/2002, the General Food Law, was enacted to provide the framework for European food safety legislation (European Parliament and Council, 2002).

The regulation rests on seven principal components: corporate responsibility across the food supply chain, traceability of raw materials and final products, risk-based official controls grounded in the precautionary principle, independent scientific risk assessment institutionally supported by the European Food Safety Authority (EFSA), a formal separation between scientific risk evaluation and political risk management, and transparent risk communication among operators, regulators, and the public.

1.3.1 The institutional architecture: EFSA and the Rapid Alert System for Food and Feed (RASFF)

EFSA, based in Parma, conducts independent scientific risk assessments for food and feed safety in the EU (European Food Safety Authority, 2023), evaluating experimental and epidemiological data to establish tolerable intake thresholds and

advise on maximum contaminant levels. The Rapid Alert System for Food and Feed (RASFF), administered under the same Regulation (European Commission, 2023a), disseminates food safety risk information among Member States, the Commission, and EFSA, offering an external benchmark against which laboratory-derived risk intelligence can be verified.

1.3.2 From Regulation 466/2001 to Regulation (EU) 2023/915

Council Regulation (EEC) No 315/93 introduced the ALARA (As Low As Reasonably Achievable) principle for contaminant management (Council of the European Communities, 1993). Commission Regulation (EC) No 466/2001 then consolidated maximum contaminant levels into one document (European Commission, 2001), superseded by Regulation (EC) No 1881/2006 (European Commission, 2006), which accumulated more than 37 amendments and roughly 75 annex footnotes over 17 years.

Commission Regulation (EU) 2023/915 took effect on 25 May 2023, replacing Regulation 1881/2006 in its entirety (European Commission, 2023b) and reorganizing regulated contaminants into six thematic sections:

1. Mycotoxins;
2. Natural plant toxins;
3. Heavy metals and inorganic contaminants;
4. Persistent organic pollutants and per- and polyfluoroalkyl substances (PFAS);
5. Process contaminants;
6. Nitrates and other environmental contaminants.

This section-based framework substantially informs the analytical organization of the pipeline discussed in this thesis, and the previous 75-footnote system was eliminated in favor of a focused observations column, leaving only 15 residual notes. Table 1.1 outlines the principal differences between the two frameworks. Among the amendments with direct operational implications for laboratories, PFAS received maximum permissible levels for the first time, following EFSA's 2020 finding that dietary exposure already exceeds the group tolerable weekly intake for a significant share of the European population (EFSA Panel on Contaminants in the Food Chain, 2020a; European Commission, 2022); pyrrolizidine alkaloid limits were broadened in parallel (European Commission, 2020), and the dilution/detoxification ban was extended from mycotoxins alone to all regulated categories.

Table 1.1: Comparative overview of Regulation (EC) No 1881/2006 and Regulation (EU) 2023/915.

Aspect	Reg. 1881/2006	Reg. 2023/915
Status	Repealed 25 May 2023	In force
Amendments	>37	Living instrument
Structure	Unstructured annex	6 thematic sections
Footnotes	~75	15 + observations column
Key definitions	Absent	Explicitly included
Detoxification ban	Mycotoxins only	All contaminant categories
PFAS limits	Absent	Included
Pyrrolizidine alk.	Absent	Included
Glycidyl esters	Absent	Included
Ergot alkaloids	Absent	Included
National derogations	Absent	Present (Art. 7)

1.4 Chemical contaminants in food

According to Regulation (EEC) No 315/93, chemical contaminants are substances not intentionally added to food, arising from environmental contamination, agricultural practices, industrial processing, storage conditions, or technological transformations during food production. Occurrence patterns, toxicological characteristics, and regulatory significance differ widely across the six thematic sections introduced above, summarised here at the level needed for the chapters that follow.

- **Mycotoxins** (Section 1): fungal metabolites of *Aspergillus*, *Fusarium*, and *Penicillium*, mainly in cereals and oilseeds. Aflatoxin B1 is an IARC Group 1 carcinogen with the strictest limits; fumonisins B1 and B2 are regulated as a combined total (Battilani et al., 2016).
- **Natural plant toxins** (Section 2): tropane alkaloids from co-harvested weeds, and pyrrolizidine alkaloids in honey and herbal teas, genotoxic with no safe threshold (EFSA Panel on Contaminants in the Food Chain, 2016).
- **Heavy metals and inorganic contaminants** (Section 3): lead, cadmium, mercury, and inorganic arsenic, none biodegradable or detoxifiable once in the food matrix (EFSA Panel on Contaminants in the Food Chain, 2010, 2012a,b). Dehydrated matrices concentrate residual metal content and are corrected for explicitly in NB1 (Section 2.3).
- **Persistent organic pollutants and PFAS** (Section 4): dioxins, dioxin-like PCBs, and PFAS, lipophilic and bioaccumulative (EFSA Panel on Contaminants in the Food Chain, 2018, 2020a). EFSA's 2020 opinion found that dietary

PFAS exposure already exceeds the group tolerable weekly intake for much of the European population.

- **Process contaminants** (Section 5): acrylamide, glycidyl esters, polycyclic aromatic hydrocarbons, and 3-MCPD, all formed during thermal processing.
- **Nitrates and other environmental contaminants** (Section 6): inorganic nitrates and perchlorates, of largely agronomic rather than supply-chain origin (EFSA Panel on Contaminants in the Food Chain, 2017, 2020b). Neither this class nor process contaminants returns a HIGH-tier signal in the DSS results reported in Chapter 3.

The chapter so far has treated food safety surveillance from its industrial side: compliance records, regulatory thresholds, laboratory throughput. The section that follows treats it from its biological side: shared molecular targets, gene networks, pathway convergence. Neither view is sufficient alone. A model that ranks contaminant–submatrix pairs by statistical and machine-learning evidence answers only where non-conformity clusters historically; it says nothing about why structurally unrelated contaminants might act as a combined threat at the molecular level. A gene-pathway network built without reference to which pairs a laboratory actually flags as high-risk, conversely, has no way of prioritising which mechanisms are operationally relevant. Bioinformatics, in the sense used throughout this thesis, treats both as instances of the same underlying problem: large, heterogeneous, structured data, laboratory compliance records in one case, curated chemical-gene interactions in the other, that only becomes interpretable once organised, modelled, and mined computationally. Chapters 2 and 3 bring the two sides back together in a single Decision Support System.

1.5 The One Health perspective

WHO, FAO, the World Organisation for Animal Health, and the United Nations Environment Programme jointly set out the One Health framework in the One Health Joint Plan of Action 2022–2026 (World Health Organization and Food and Agriculture Organization and World Organisation for Animal Health and United Nations Environment Programme, 2022): human, animal, and environmental health are seamlessly linked, and progress in one domain cannot be sustained without integrated attention to the others. Applied to food contamination surveillance, this brings into view risk facets that traditional regulatory systems miss, since those systems judge isolated contaminants against separate maximum permissible levels.

Contaminants introduced into the food chain seldom stay confined to a single environmental compartment: agricultural soil pollutants, industrial emissions, and polluted irrigation water act on crops, livestock feed, aquatic systems, and wildlife habitats at once, bioaccumulating in animal tissues and environmental sinks. Food safety surveillance data acquired by accredited laboratories therefore function as indirect epidemiological indicators of broader ecosystem health, with relevance well beyond simple compliance questions tied to individual products.

1.5.1 Shared molecular targets and combinatorial toxicity

A One Health perspective in food contaminant monitoring means looking beyond individual toxicological profiles to the molecular processes through which structurally varied contaminants converge on common biological sites, in keeping with a systems toxicology approach to risk assessment (Sturla et al., 2014). In practical dietary scenarios, people encounter complex chemical mixtures rather than isolated compounds, and the biological implications of simultaneous exposure may diverge markedly from the additive effects of single compounds. Where multiple contaminants vie for identical efflux transporters, inhibit shared detoxification enzymes, or activate mutual nuclear receptors, aggregated biological impacts can surpass what individual assessments would predict, even when each contaminant remains below its own regulatory threshold, a fundamental oversight in current food safety protocols. A structured way to reason about this synergistic cocktail effect is to ask which molecular targets are shared across contaminants in the first place: a gene interacting with many otherwise unrelated chemicals functions as a hub, and its perturbation by any one of them can alter the organism's response to all the others. Identifying such hub genes computationally, rather than inferring convergence qualitatively, is the specific bioinformatic contribution this thesis makes to the One Health side of the problem, detailed in Section 1.9.

Numerous contaminants governed by Regulation (EU) 2023/915 interact with ATP-binding cassette (ABC) transporters, which restrict intestinal absorption and tissue distribution of xenobiotics; when several substrates compete at once, cellular efflux capacity can be overwhelmed and systemic bioavailability rises collectively, a mechanism invisible to frameworks that evaluate contaminants separately. Multiple contaminants may also interact with tumor suppressor genes, steroid hormone receptors, or inflammatory signaling pathways, producing synergistic genotoxic, endocrine disrupting, or pro-inflammatory effects at exposure levels deemed individually safe. The Comparative Toxicogenomics Database (CTD) (Davis et al., 2023) makes it possible to map such chemical-gene interactions systematically and evaluate multi-contaminant convergence on shared molecular targets, the methodology at the core

of the One Health biological analysis in this thesis.

1.5.2 Food safety and planetary health

Environmental degradation and food contamination reinforce each other in both directions. Dioxins, PCBs, and heavy metals emitted from industrial activities infiltrate agricultural soil, irrigation water, and ambient air; agricultural intensification adds nitrate leaching and soil degradation, and PFAS from industrial and firefighting activities infiltrate irrigation systems in turn. Food production itself feeds back into contamination: phosphate fertilisers inherently carry cadmium from geogenic impurities, and aquaculture in polluted water concentrates contaminants in farmed fish as it absorbs industrial and agricultural runoff.

Climate change compounds this picture by altering fungal ecology, increasing drought frequency, and shifting contaminant bioavailability in soils, with toxigenic species extending into agricultural regions once considered low-risk (Battilani et al., 2016). Protecting environmental systems and preserving human health cannot really be separated, which is why food safety surveillance needs to be integrated into a broader planetary health framework, with continuous, automated systems able to flag emerging indicators before they harden into regulatory problems.

1.6 Statistical methods in food safety surveillance

Analyzing food safety monitoring data invokes unique methodological challenges. Food safety datasets, produced by accredited laboratories, are observational rather than experimental. The sampling process is shaped by client requests, regulatory requirements, and priorities based on risk assessments. These datasets end up with structural imbalances and diverse sample sizes across different groups. Distributions are rarely normal, and multiple testing issues arise too. Specialized statistical methodologies become necessary as a result. Food contaminant concentration data are typically right-skewed, often approximated by a lognormal distribution, a pattern attributed to the multiplicative nature of biological and environmental processes that govern contaminant formation, transfer, and accumulation (Limpert et al., 2001). Standard parametric methods, which assume normality, are therefore generally unsuitable for raw concentration values. Non-parametric statistical approaches are used systematically instead.

The occurrence of false positive results increases significantly when numerous statistical tests are conducted concurrently. This is common in surveillance scenarios, where

thousands of contaminant-matrix combinations undergo testing. The Benjamini–Hochberg False Discovery Rate (FDR) procedure (Benjamini and Hochberg, 1995) offers a well-balanced approach, as it manages the expected proportion of false positives among rejected hypotheses and proves less conservative than the Bonferroni correction when dealing with a large number of tests. For enrichment testing, the Fisher exact test is favored over the chi-squared approximation because it remains valid irrespective of expected cell frequencies. That property matters when numerous contaminant-matrix strata possess small sample sizes and low non-conformity counts (Fisher, 1922).

For precise interval estimation of non-conformity rates, confidence interval procedures need to remain valid even for extreme proportions and small sample sizes. The Wilson score interval (Brown et al., 2001) outperforms the classical Wald interval here. It offers substantially improved coverage properties, and it avoids mathematically invalid intervals that extend below zero or above one.

Detecting temporal trends in food contamination data is complicated by the fact that genuine secular trends, seasonal variation patterns, and irregular residual noise all occur together. The Mann–Kendall test (Mann, 1945) offers a robust non-parametric solution here, as it identifies monotonic trends without relying on distributional assumptions. In the current pipeline, this test is applied to trend components extracted through Seasonal-Trend Decomposition using LOESS (STL) (Cleveland et al., 1990). This matters because seasonal patterns, such as those driven by harvests in mycotoxins or by growing-season effects in nitrates, can otherwise obscure trend detection. Sen’s slope is used alongside it, giving a robust estimate of the trend rate that resists the influence of outliers.

1.7 Machine learning for risk classification in food safety

In the realm of food safety, machine learning approaches are increasingly utilized in predictive toxicology and epidemiological surveillance, often in tandem with the broader bioinformatic toolkit for processing large-scale biological and chemical data (Soroushianfar et al., 2025). Their appeal lies in the capacity to uncover complex, potentially non-linear relationships within large, heterogeneous datasets, relationships that traditional statistical methods often struggle to capture (Wang et al., 2022).

These methods have been applied successfully to a range of tasks. Contamination risk prediction and anomaly detection are among the most established uses. Fraud identification and exposure modeling follow closely behind, while sampling plan

optimization and the development of early warning systems round out the picture (Wang et al., 2022).

1.7.1 Binary classification and evaluation metrics

This thesis frames the prediction of food contamination risk as a supervised binary classification task, where the goal is to distinguish pairs with structurally elevated risk from those carrying routine risk. The model’s output is a calibrated probability, which yields a continuous risk score suitable for integration into the Decision Support System (DSS) scoring rule via tier assignment based on fixed thresholds. Class imbalance is a built-in feature of food safety datasets, since non-conforming analytical results are rare among observations. Traditional accuracy metrics can be deceptive here. A model that defaults to majority class prediction still achieves high accuracy, without offering any real differentiation of risk. For this reason, Precision-Recall Area Under the Curve (PR-AUC), also called average precision, is preferred over Receiver Operating Characteristic Area Under the Curve (ROC-AUC), because PR-AUC keeps the focus on the positive class and is not distorted by a large number of true negatives (Davis and Goadrich, 2006). The pipeline used here treats PR-AUC as the chief optimization metric. ROC-AUC is retained only as a secondary measure, mainly to allow alignment with existing benchmarks.

1.7.2 Gradient boosting algorithms

Ensemble learning, particularly gradient boosting, is prominently employed for exploring tabular data in machine learning applications. Gradient boosting builds its ensemble from weak learners, typically shallow decision trees. Each new learner is optimized to address the residuals left by its predecessors, so prediction error is minimized step by step (Friedman, 2001). The method is valued for its adaptability and its regularization capabilities, which counteract overfitting. It also handles complex feature interactions well.

Random Forest (Breiman, 2001), used here as the baseline comparator, is an ensemble of decision trees built through bagging with random feature subsampling.

It is not a gradient boosting technique. Its main difficulty, instead, lies in the need for explicit feature encoding strategies when categorical spaces have high cardinality.

XGBoost (Chen and Guestrin, 2016) builds on gradient boosting, adding histogram-based tree construction and L_1/L_2 regularization. It also handles missing values well. The `eval_metric` parameter is what ties the internal optimization objective to the external evaluation metric, keeping the two aligned throughout training.

LightGBM (Ke et al., 2017) speeds up convergence through leaf-wise tree growth, and it performs efficiently on extensive datasets. Class imbalance is handled through its `is_unbalance` parameter.

CatBoost (Prokhorenkova et al., 2018) was developed to efficiently manage high-cardinality categorical variables, and it employs Ordered Target Encoding to do so. This method overcomes the target leakage associated with conventional target encoding, where category mean computation on full datasets inadvertently embeds label information. CatBoost’s ordered encoding instead performs statistical computation for each sample based solely on preceding samples in a random permutation. Leakage is eliminated this way, yet essential category frequency statistics are still preserved.

Food safety datasets often contain numerous distinct food submatrices and contaminant categories. In such settings, this capability proves advantageous, and it helps explain CatBoost’s exemplary performance in similar machine learning benchmarks for food safety (van den Bulk et al., 2025).

1.7.3 Model interpretability and SHAP values

Predictive accuracy alone is not enough for a model to be useful. Stakeholders, clients, regulatory bodies, and academic peers among them, need model outputs that are transparent, auditable, and scientifically explicable. Stakeholders simply cannot vouch for the legitimacy of a risk evaluation, or work out what is actually driving a prediction, if the model stays a black box.

SHAP (SHapley Additive exPlanations) (Lundberg and Lee, 2017) offers a theoretically grounded way to break model predictions down into additive feature contributions, built on Shapley values from cooperative game theory. Each feature’s contribution is calculated as its mean marginal contribution across all possible permutations of features. This is what guarantees a fair allocation of responsibility among them. Three axioms give SHAP values their mathematical footing. The efficiency axiom ensures that feature contributions sum to the prediction itself. The symmetry axiom requires equal values for features with identical marginal contributions. The dummy axiom assigns zero contribution to any feature with no real influence on predictions. For tree-based models specifically, TreeExplainer computes SHAP values both precisely and efficiently. It supports global assessment of feature importance as well as local explanation of individual predictions.

1.8 Decision support systems in food safety

In food safety, a Decision Support System (DSS) functions as a computational mechanism designed to bolster structured decisions by domain experts, drawing together diverse data sources, quantitative models, and visual interfaces (Sprague, 1980). The guiding principle of DSS design is not to replace expert judgment but to sharpen it. Computational tools structure and quantify what would otherwise remain scattered, lending insights an evidentiary support that experts could not easily assemble, or manage, on their own.

The food safety surveillance landscape in the European Union requires every food business operator to develop a risk-based analytical self-monitoring plan, built into their Hazard Analysis and Critical Control Points (HACCP) system, as stipulated by Regulation (EC) No 852/2004 (European Parliament and Council, 2004).

The regulatory framework sets out maximum levels for evaluating analytical results, but it says nothing about how to choose which contaminant-matrix combinations to test, how often to sample, or how to rank priorities. In practice, these decisions rest on a mix of sources, including expert judgment, RASFF alert histories, EFSA occurrence data, sector-specific guides, and whatever historical testing data has accumulated over time. The approach is hard to scale. It is also vulnerable to information gaps, particularly for novel contaminants or unusual ingredient and source combinations, and it consumes a considerable amount of specialist time. A laboratory-based DSS that combines proprietary occurrence data with current regulatory maximum levels addresses this shortfall effectively, and doing so at the scale of a full laboratory portfolio is precisely the kind of structured-data problem bioinformatic pipelines are built to handle.

Public databases, including RASFF, EFSA occurrence data, and national monitoring program reports among them, simply cannot match the granularity, timeliness, and market-specific precision that data from a high-throughput accredited laboratory can offer. The DSS developed in this thesis draws on exactly this kind of proprietary resource. It generates three tiers of actionable risk intelligence, namely high, medium, and low, derived from statistical enrichment evidence combined with machine learning risk probabilities. The result is a shift away from reactive compliance testing toward proactive, data-driven risk intelligence.

The scoring rule behind the Decision Support System (DSS) was built with a specific tension in mind. That tension is the trade-off between two error types that dominate food safety monitoring. Consider a false negative first. Assigning a low tier to what is actually a high-risk pair can let contamination go undetected, and from there the consequences cascade, including market withdrawals and adverse effects on

consumer health. A false positive works the other way, a high tier attached to a pair that is genuinely low risk, and the cost here is different, mainly wasted monitoring spend. The DSS handles this through dual-evidence AND/OR logic. Its aim is to hold down false positives at the tier where operational cost is highest, while still catching emerging signals, that is, keeping recall strong where it matters.

1.9 The One Health biological framework: CTD, KEGG, and network analysis

1.9.1 The Comparative Toxicogenomics Database

The CTD (Davis et al., 2023), curated at Mount Desert Island Biological Laboratory, is a manually curated public resource. It integrates data on chemical-gene, chemical-disease, and gene-disease interactions, extracted with care from peer-reviewed biomedical literature. Every record undergoes individual review by a trained curator, who validates experimental evidence, categorizes the interaction type using a controlled vocabulary, and assigns confidence levels. The result is a dataset with markedly higher precision than automated text mining alternatives can offer.

For categorizing interaction types, the CTD relies on a structured vocabulary. It covers expression modulation, binding effects, methylation influence, phosphorylation changes, and activity impacts, among other categories. This lets network analysis distinguish transcriptional regulation from post-translational modification and from direct molecular binding. The distinction matters for food safety work, since whether a contaminant activates or inhibits a gene carries direct toxicological consequences, so specificity here is not a minor detail.

CTD's chemical coverage includes all contaminant classes governed under Regulation (EU) 2023/915, among them mycotoxins, heavy metals, persistent organic pollutants, and PFAS, alongside process contaminants and natural plant toxins. Given this breadth, CTD stands as the primary resource for chemical-gene network analysis oriented towards food safety. Protein-protein interaction databases like STRING (Search Tool for the Retrieval of Interacting Genes/Proteins) (Szklarczyk et al., 2019) serve a different purpose. Functional annotation resources such as Gene Ontology (Gene Ontology Consortium, 2000) do not specifically address chemical exposure conditions. And the Reactome pathway database is oriented toward normal cellular processes, not the perturbations that toxicants induce.

1.9.2 KEGG pathway enrichment analysis

The Kyoto Encyclopedia of Genes and Genomes (KEGG) (Kanehisa and Goto, 2000) provides curated mappings of metabolic pathways, signalling cascades, and disease-associated biological systems, built on systematic gene function analyses across different organisms. This resource proves pivotal for conducting KEGG pathway enrichment analysis, a method that tests whether a particular set of genes associated with food contaminants through CTD interactions is overrepresented in specific pathways compared to random genomic sampling. KEGG was chosen as the primary reference for pathway analysis here because its structured, curated pathway maps encode known metabolic, detoxification, and disease-signalling systems. These map directly onto the concerns of toxicological assessment. Enrichment analysis is also run against Gene Ontology Biological Process and Gene Ontology Molecular Function, alongside Reactome gene set libraries, using the Enrichr API (Kuleshov et al., 2016). Together these give a multifaceted view of biological function.

Pathway enrichment analysis matters because it turns a raw list of gene targets into a biologically interpretable framework. It does more than note that a contaminant portfolio hits a given set of genes. It shows where these genes cluster disproportionately, namely in lipid metabolism, in carcinogenesis, in ABC transporter activity, in inflammatory signaling. The molecular interactions gain meaning here. They connect to biological processes that matter directly to toxicology and clinical practice.

1.9.3 Biological network analysis

Network biology, drawing on graph theory, offers a way to represent and analyze biological interaction systems (Barabasi and Oltvai, 2004). Nodes stand for genes, proteins, pathways, or contaminants. Edges capture the biological relationships that connect them. Complex systems become depictable this way. Properties emerge from these networks that isolated examination of individual components simply cannot reveal.

Central to network biology is the concept of a hub node, a node characterized by significantly higher connectivity than the network average.

Scale-free biological networks, typified by a few highly connected nodes and many with minimal connections, assign critical functional roles to these hubs. Disrupting a hub gene through chemical exposure can affect every associated biological process. The effects can cascade across multiple molecular pathways. Hub genes within a chemical-gene network therefore act as molecular bottlenecks for combinatorial toxicity, since perturbation by one contaminant can modify cellular responses to

other contaminants targeting the same gene. The tripartite architecture outlined in this thesis links contaminant, gene, and pathway nodes. It supports visualization of contaminant-gene interactions, and it allows hub genes shared among multiple contaminants to be identified. It also enables systematic perturbation of biological pathways across the contaminant portfolio to be examined. This systems-level perspective provides mechanistic biological context. In doing so, it transforms statistical and machine learning risk assessments from empirical observations into conclusions that are scientifically interpretable and defensible to regulators. The specific software implementation of this network analysis is described in Chapter 2.

1.10 Aims of the thesis

This thesis develops, validates, and applies a laboratory-based Decision Support System for assessing food contaminant risks. The system draws on statistical inference and machine learning. It also incorporates One Health biological network analysis. Historical analytical records generated by Neutron Laboratories, produced pursuant to Regulation (EU) 2023/915, inform this work.

Framed this way, the thesis brings together two bioinformatics traditions that are not usually combined in the same pipeline. The first is the statistical and machine learning treatment of large-scale laboratory records, the same class of methods bioinformatics applies to any high-dimensional structured dataset, used here to convert raw compliance data into a leakage-free risk classifier (Chapters 2–3). The second is the more familiar bioinformatics territory of gene-pathway network analysis, built from the Comparative Toxicogenomics Database and KEGG, used here to ground the DSS output in molecular mechanism (Section 2.7). Neither layer is bioinformatic merely by virtue of touching biological data; it is the integration of the two, statistical and machine learning evidence converging with biological network evidence on the same set of contaminants, that constitutes the thesis’s bioinformatic contribution, a point returned to explicitly in the concluding chapter.

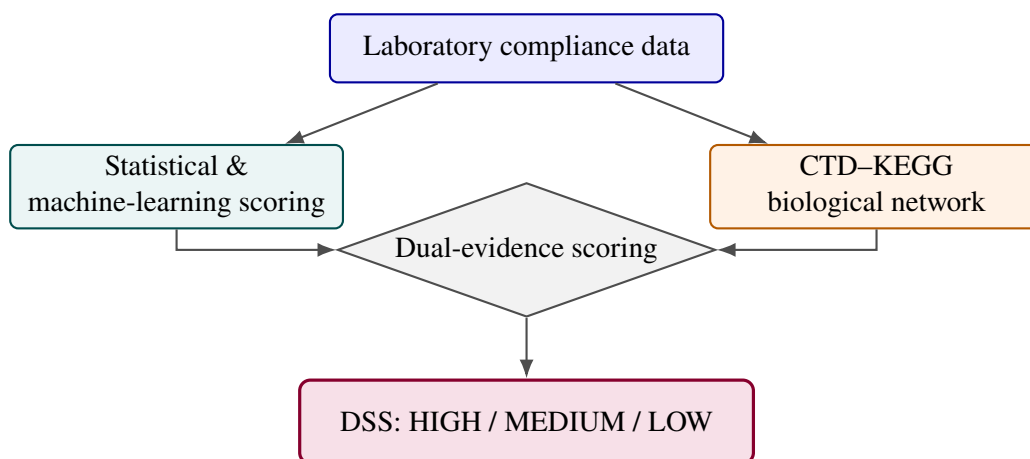


Figure 1.1: Conceptual overview of the thesis: two independent evidence streams converge in the DSS.

The specific goals are as follows:

- Transforming routine laboratory compliance data into epidemiological risk intelligence;
- Identifying statistically enriched contaminant-matrix combinations via Fisher exact testing combined with Benjamini-Hochberg False Discovery Rate correction;
- Creating machine learning classifiers capable of predicting elevated contamination risk from topological features free of leakage;
- Assessing temporal contamination dynamics through Seasonal-Trend Decomposition and non-parametric Mann-Kendall trend analysis;
- Providing biological context to contaminant interactions through CTD-KEGG network analysis under a One Health framework, focusing on identifying hub genes and shared pathways that mediate combinatorial toxicity;
- Implementing a fully automated, Docker-containerized, and Papermill-orchestrated pipeline, designed for operational deployment in an accredited laboratory setting.

Chapter 2

Materials and Methods

2.1 Study Design and Computational Framework

This work took shape as a retrospective observational study, one that brings SQL-based data engineering, exploratory statistical analysis, enrichment testing, machine learning classification, dual-evidence decision scoring, and biological network analysis together into a single Decision Support System (DSS) for food contaminant risk assessment within an ISO/IEC 17025 (International Organization for Standardization, 2017) accredited laboratory environment. Every stage of this pipeline operates on the same unit of analysis, a contaminant–submatrix pair, such as aflatoxin B1 tested in maize germ; the pipeline’s end product is a risk tier, HIGH, MEDIUM, LOW, or INSUFFICIENT, assigned to each such pair (Section 2.4 defines both terms formally and gives the full parameter glossary). The analytical workflow followed a six-stage modular pipeline, conventionally labelled NB1 through NB4 with two additional validation stages. Each stage reads a documented input file produced by the stage before it and writes a documented output file consumed by the stage after it. This separation means any single stage can be re-run, audited, or replaced without disturbing the rest of the pipeline. It also means the provenance of every downstream number can be traced back to a specific, inspectable intermediate file. All analyses were carried out on a workstation running Microsoft Windows 11. Python 3.10 was used throughout the data engineering, statistical analysis, and machine learning stages (NB1 through NB3b), while R 4.5 handled the biological network analysis in NB4. For this class of problem, the igraph (Csardi and Nepusz, 2006) and ggraph ecosystem for network construction and visualisation is considerably more mature than its Python equivalents.

The pipeline is orchestrated end to end through Papermill (nteract contributors, 2018), which executes each notebook programmatically by injecting runtime parameters from

a single centralised configuration file, `config.yml`, into a designated parameters cell at the top of each notebook. Under this design, no numerical threshold, file path, or operational switch is embedded directly in the notebook code. Every threshold quoted later in this chapter is read from, and in some cases written back into, this one configuration file. Executed notebooks are archived with execution timestamps to preserve a complete audit trail. A checkpoint mechanism also verifies the existence of each stage’s primary output file before the next stage is allowed to run. The complete pipeline is containerised in a Docker (Merkel, 2014) image bundling Python 3.10, R 4.5, and every required library. This allows the same pipeline to be executed identically on any compatible machine. Table 2.1 summarises the complete software inventory and the role each tool plays at each stage.

Table 2.1: Software environment: tools, libraries, and pipeline roles.

Tool / Library	Category	Purpose	Stage
SQL Server	Database	LIMS extraction	NB1–NB2
Python 3.10	Language	Core pipeline	NB1–NB3b
R 4.5	Language	Network biology	NB4
pandas / numpy	Wrangling	Cleaning, aggregation	NB1–NB3
scipy / statsmodels	Statistics	Fisher, Wilson, STL	NB1–NB2
pymannkendall	Statistics	Mann–Kendall testing	NB2
scikit-learn	ML framework	Cross-validation, metrics	NB3
CatBoost	ML model	Primary classifier	NB3
XGBoost / LightGBM / RF	ML model	Benchmark models	NB3
shap	Interpretability	Feature attribution	NB3
igraph / ggraph	Network	Tripartite graph	NB4
CTD REST API	External	Chemical–gene data	NB4
gseapy / Enrichr	Enrichment	KEGG, GO, Reactome	NB4
Papermill	Automation	Notebook orchestration	Pipeline
Docker	Container	Reproducibility	Pipeline
Streamlit	Dashboard	Live operator interface	Deployment
HTML export script	Dashboard	Offline standalone report	Deployment
Power BI	Dashboard	Executive reporting	Deployment

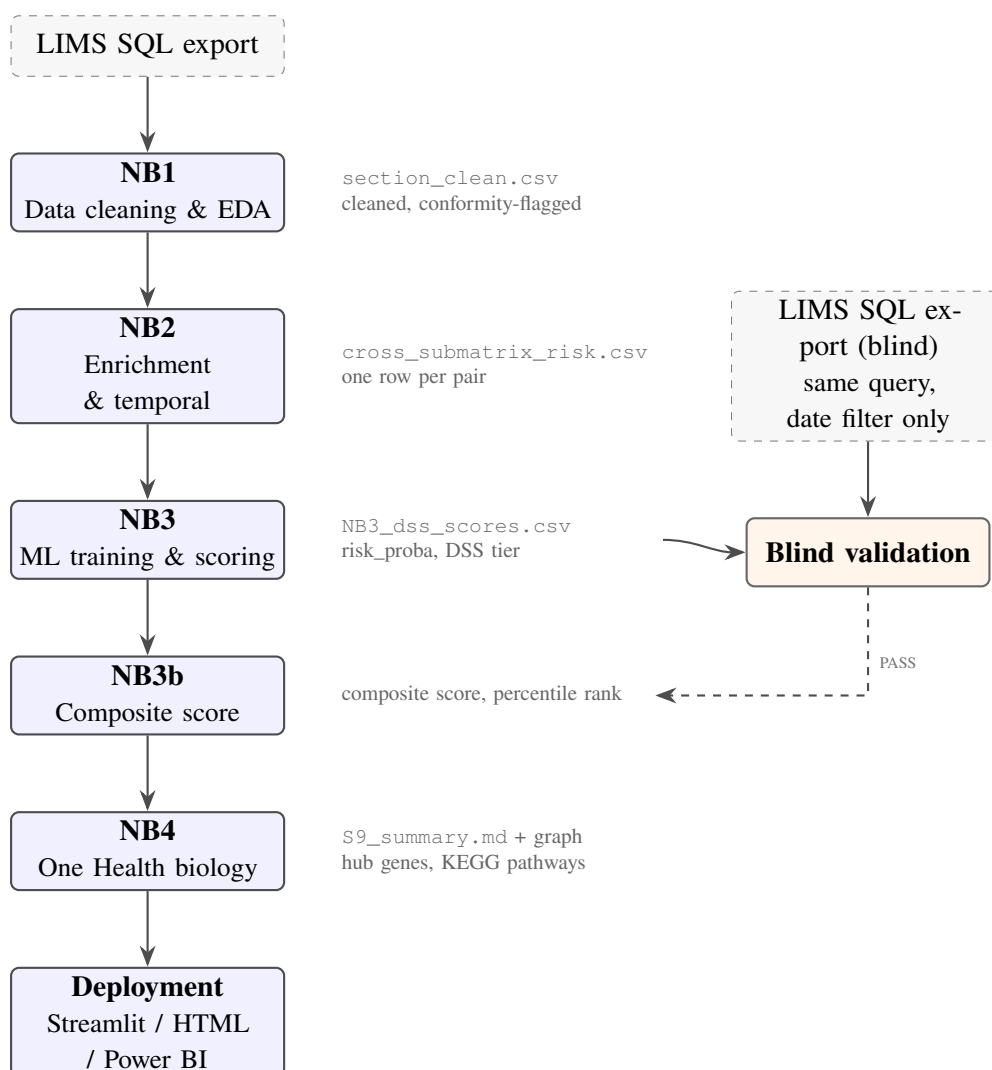


Figure 2.1: Pipeline architecture. Each stage reads the output file of the preceding stage, with the content of every intermediate file summarised alongside it; full field definitions are given in Table 2.2 and in the sections that follow. The frozen model and DSS thresholds produced by NB3 are applied unchanged to both blind validation windows; the dashed arrow indicates that the blind validation result informs interpretation but does not modify any upstream parameter.

2.2 Data Source: LIMS Architecture and Extraction

All analytical records originate from the Neutron Laboratories LIMS, which is hosted on a Microsoft SQL Server instance and accessed through a read-only credential, since the pipeline is designed never to write back to the production database. The LIMS follows a star-schema relational architecture, where a single fact table stores one certified analytical result per row. This fact table is joined against dimension tables that provide the identity of the contaminant, its mapping onto one of the six thematic sections of Regulation (EU) 2023/915, a two-tier food submatrix hierarchy, sample metadata, analytical method codes, and measurement units. Where a state-specific regulatory limit exists for a given contaminant–matrix combination, it takes

precedence. If none exists, the extraction query falls back to the generic limit defined at the section level. Every row therefore carries an unambiguous applicable maximum level at extraction time, and no separate lookup is required downstream.

The training extraction filters this LIMS to the window of January 2024 through December 2025, twenty-four consecutive months, and returns all certified analytical results spanning the six regulatory sections of Regulation (EU) 2023/915. Two full calendar years give sufficient depth for the seasonal-trend decomposition and Mann–Kendall trend analysis described later in this chapter. The window’s start also aligns with the entry into force of Regulation (EU) 2023/915, and twenty-four months ensures that all six regulatory sections are represented with enough volume to support section-stratified statistical testing. Two further extractions were performed for the purpose of blind validation, using the identical query with no modification beyond the date filter, covering a prospective window from January through May 2026 and a retrospective window covering the whole of calendar year 2023. Section 2.8 develops the rationale for evaluating in both temporal directions.

2.3 Data Preparation and Exploratory Analysis (NB1)

2.3.1 Data Cleaning

Raw analytical values in the LIMS are stored as free-text strings rather than as clean numeric fields. The first task of NB1 is therefore a deterministic parsing pass that converts this free text into reliable numbers. Confirmed non-detections, recorded as “N.R.”, are converted to a numerical zero and retained as valid conforming observations. Entries below the analytical method’s limit of quantification, recorded in the format `<1q(val)`, are dropped from both the non-conformity count and the total valid count. They represent left-censored values, and including them would introduce a systematic bias. Administrative strings and internal quality-control markers are discarded outright.

Two further corrections are applied. Values reported using the European decimal-comma convention are normalised to decimal-point notation before numeric conversion. For dehydrated plant-derived matrices such as dried herbs, algae, mushrooms, and vegetable powders, the applicable regulatory maximum level is multiplied by a dry-matter concentration factor of five before the conformity check is applied. This step reflects the physical consequence of water removal on the residual mineral content, so that the non-conformity flag correctly accounts for the concentration effect of drying. The industrial tutor confirmed this correction factor against established laboratory practice.

Regulatory limit strings are resolved through a cascaded strategy. Direct numeric conversion is attempted first. If that fails, the parser falls back on a lookup in a hand-curated dictionary of contaminant-specific entries covering composite parameters such as the summed aflatoxins limit. A generalised regular-expression extraction is used only as a last resort.

2.3.2 Exploratory Data Analysis

NB1 goes on to characterise the distributional properties, temporal dynamics, and food matrix structure of the contaminant data once cleaning is complete, treating each of the six regulatory sections separately. Descriptive statistics are computed at both the contaminant level and the food matrix level for every stratum with at least five observations. Normality is tested first, using the Shapiro–Wilk test for strata of five thousand observations or fewer and the Anderson–Darling test for larger strata. Contaminant concentration data are well known to follow a right-skewed, approximately lognormal distribution (Limpert et al., 2001). This departure from normality is what formally justifies relying on non-parametric methods throughout the pipeline.

Where concentration differences across food matrices need to be assessed, NB1 applies the Kruskal–Wallis test,

$$H = \frac{12}{N(N+1)} \sum_{i=1}^k \frac{R_i^2}{n_i} - 3(N+1), \quad (2.1)$$

where N is the total number of observations pooled across all groups, k is the number of food matrix groups being compared, n_i is the size of group i , and R_i is the sum of the ranks assigned to group i . Where the omnibus test indicates a significant difference, pairwise Mann–Whitney U post-hoc tests with Bonferroni correction identify which specific pairs of matrices differ, and Cohen’s d quantifies the effect size in standardised units,

$$d = \frac{|\bar{x}_A - \bar{x}_B|}{s_{\text{pooled}}}, \quad s_{\text{pooled}} = \sqrt{\frac{(n_A - 1)s_A^2 + (n_B - 1)s_B^2}{n_A + n_B - 2}}. \quad (2.2)$$

Detecting genuine temporal trends in this kind of data is complicated by strong harvest-driven seasonal cycles, and for this reason NB1 applies a two-stage approach. Where at least twenty-four monthly observations are available, the series is first decomposed using Seasonal-Trend Decomposition based on LOESS (STL),

$$Y_t = T_t + S_t + R_t, \quad (2.3)$$

which separates the observed monthly value Y_t into a trend component T_t , a seasonal component S_t , and a residual component R_t . The Mann–Kendall trend test is then applied to the isolated trend component T_t alone, so that a genuine secular increase is not masked by recurring seasonal peaks. Where fewer than twenty-four monthly observations are available, decomposition is not attempted and the Mann–Kendall test is applied directly to the raw monthly means. The Mann–Kendall statistic is

$$S = \sum_{i < j} \text{sgn}(x_j - x_i), \quad \tau = \frac{S}{n(n-1)/2}, \quad \tau \in [-1, +1], \quad (2.4)$$

and where a trend is detected, Sen’s slope (Sen, 1968) provides a robust, outlier-resistant estimate of its rate,

$$\hat{\beta} = \text{median} \left\{ \frac{x_j - x_i}{j - i} \mid j > i \right\}, \quad (2.5)$$

reported in concentration units per month. Outliers are flagged using Tukey’s fence, $Q_3 + 1.5 \times \text{IQR}$, but are never removed from the dataset, since in this context an extreme value is far more likely to represent a genuinely contaminated batch than a measurement artefact. Wilson 95% confidence intervals (Wilson, 1927; Brown et al., 2001) are also computed at the submatrix level for every stratum with at least five observations; their formula is given in Section 2.5 where they are used at pair level.

2.4 Pipeline configuration and parameter glossary

The unit of analysis throughout this pipeline is not a single contaminant, but a contaminant–submatrix pair: for example, aflatoxin B1 tested specifically in maize germ, as distinct from aflatoxin B1 tested in soy. Every statistic, model score, and risk tier reported in this thesis is computed per pair, because the same contaminant can behave very differently across food matrices, and collapsing it into a single value would hide exactly that variation. Section 2.2 describes how each pair is extracted from the LIMS; Section 2.5 defines the statistics computed for it; Section 2.6 describes how a single classifier, not one model per contaminant, is trained across all pairs simultaneously, with contaminant and submatrix identity entering only as two of its eight input features (Section 2.6.2).

Two counts recur throughout this chapter and are worth fixing before anything else:

for a given pair, n_{total} is the number of valid analytical results recorded for it, and n_{nc} is how many of those results exceeded the applicable regulatory maximum level, i.e. were non-conforming. Both are properties of the pair alone, computed directly from the LIMS extract, and every threshold introduced below is defined in terms of one or the other.

Because thousands of such pairs are tested independently within each regulatory section (Section 2.5), a multiple-testing correction is required to control the rate of spurious findings that would otherwise arise from running so many statistical tests in parallel; the Benjamini–Hochberg false discovery rate procedure, detailed in Section 2.5, serves this purpose.

All operational thresholds are centralised in `config.yml`, so a single file governs behaviour that would otherwise be scattered across scripts, and every value quoted later in this chapter is read from, and in some cases written back into, that one file. A fixed random seed of 42 is passed to every stochastic operation in the pipeline, from the train/test split to the cross-validation folds to the gradient boosting models themselves.

Table 2.2 collects every fixed parameter used from this point on. It is meant to work as a reading aid rather than as a self-contained explanation: each notebook section that follows (NB1 through NB4) introduces its own parameters in context, with the full reasoning behind each value, and this table exists only so that a name like `MIN_NC` or `DSS_HIGH` can be looked up in one place the moment it resurfaces later, rather than requiring the reader to search back through the chapter for where it was first defined.

Table 2.2: Parameter glossary. Full rationale and formal definitions are given in the referenced section.

Parameter	Value	Section	Meaning
Pair	contam. $\times$ submatrix	2.2	unit of analysis throughout: one contaminant in one sub- matrix
$n_{\text{total}} / n_{\text{nc}}$	–	2.2	total results / non-conforming results for a pair, from the LIMS
MIN_N (en- richm.)	5	2.5	min. n_{total} for a pair to enter NB2 enrichment testing
MIN_N (model.)	15	2.6	min. n_{total} for a pair to enter NB3 model training
TARGET	0 / 1	2.6	binary label a pair gets for train- ing: 1 if $n_{\text{nc}} \geq \text{MIN_NC}$
MIN_NC	1	2.6	min. n_{nc} for TARGET = 1, Eq. 2.10
BH α	0.05	2.5	false discovery rate threshold, applied per regulatory section
DSS_HIGH / DSS_MED	90th / 75th pct.	2.5	DSS tier boundaries, set on the training non-conformity rate distribution
Cutoff	0.50	2.6.6	CatBoost probability above which a pair counts as model- positive for DSS scoring
Seed	42	2.4	fixed random seed, every stochastic step

Applying two different values of MIN_N at two separate stages was not an oversight but a deliberate methodological choice. NB2 uses the more permissive threshold of five observations because its purpose is exploratory: it characterises the full landscape of contaminant–submatrix combinations, including rare ones, and the DSS tier boundaries are later derived from that full landscape, so narrowing it artificially at this stage would bias the very thresholds the system depends on. NB3 is different in kind because it trains a supervised classifier, and raising the bar to fifteen observations before a pair is allowed to contribute to training directly addresses the risk of fitting noise on very small samples. Whether fifteen was the right choice, rather than some other value, is examined in the sensitivity analysis reported in the results chapter.

2.5 Statistical Enrichment and Temporal Analysis (NB2)

NB2 works from the cleaned section-level data produced by NB1 and carries out all statistical inference at the finer granularity of individual contaminant–submatrix

pairs. The minimum support threshold of $\text{MIN_N} = 5$ applies here as well. Every enrichment test and every multiple-testing correction is run independently within each of the six regulatory sections, rather than pooled across the whole dataset. This section-by-section stratification is necessary because baseline non-conformity rates differ substantially across sections. Pooling the tests would mean comparing a pair against a background rate that is largely irrelevant to it.

For every pair, NB2 computes the total number of valid observations n_{total} , the number of non-conforming observations n_{nc} , and the resulting non-conformity rate $\text{nc_rate} = n_{\text{nc}}/n_{\text{total}}$. It also computes the monthly coefficient of variation of the analytical count series,

$$\text{cv_monthly} = \frac{\sigma_{\text{monthly}}}{\mu_{\text{monthly}}}, \quad (2.6)$$

computed entirely from how often a pair was sampled month by month, with no reference to how many of those samples were non-conforming; this makes it safe to use as a machine learning feature without any risk of leaking information about the target.

For each pair with at least five total observations, a one-sided Fisher exact test evaluates whether that specific contaminant–submatrix cell’s non-conformity rate exceeds the background rate of the same contaminant across every other submatrix within the same section. The null hypothesis states that the odds ratio equals one,

$$\text{OR} = \frac{n_{\text{nc}} \cdot d}{(n_{\text{total}} - n_{\text{nc}}) \cdot c}, \quad (2.7)$$

where c and d are the remaining two cells of the 2×2 contingency table, built from every other submatrix in which that same contaminant was tested within the same section: $c = n_{\text{nc}}^{(\text{contam}, \text{rest})}$, the non-conformity count for that contaminant across its other submatrices in the section, and $d = n_{\text{total}}^{(\text{contam}, \text{rest})} - n_{\text{nc}}^{(\text{contam}, \text{rest})}$. This choice of background makes the test more specific: it asks whether a given submatrix is unusually problematic for a contaminant that otherwise behaves normally, rather than whether a combination merely sits above the section average. The Fisher exact test was chosen over the chi-squared approximation because chi-squared requires an expected count of at least five in every cell, a condition that fails routinely for low-prevalence strata in this dataset.

Raw p -values are corrected within each section using the Benjamini–Hochberg false discovery rate procedure,

$$k^* = \max \left\{ k : p_{(k)} \leq \frac{k}{m} \alpha \right\}, \quad (2.8)$$

and only pairs satisfying $q < 0.05$ under this correction are retained as significantly enriched. Benjamini–Hochberg was preferred over Bonferroni because the latter would suppress the majority of genuine signals along with false ones given the volume of pairs tested.

Every pair-level non-conformity rate is reported alongside a Wilson 95% confidence interval (Wilson, 1927; Brown et al., 2001),

$$CI_W = \frac{n_{nc} + \frac{z^2}{2} \pm z \sqrt{\frac{n_{nc}(n_{total} - n_{nc})}{n_{total}} + \frac{z^2}{4}}}{n_{total} + z^2}, \quad z = 1.96, \quad (2.9)$$

chosen in preference to the classical Wald interval because Wilson intervals remain well-behaved even for extreme proportions and small samples.

At the final substep of NB2, the two DSS tier boundaries come from the empirical distribution of submatrix-level mean non-conformity rates across all pairs with at least five observations. DSS_HIGH is set at the 90th percentile of that distribution, while DSS_MED sits at the 75th percentile. Both values are computed once, on the training data only, then frozen and never recalculated for any subsequent blind evaluation: if the proportion of high-risk pairs appears to shift under a frozen threshold, that shift is itself diagnostic information about how the underlying data has moved, and letting the threshold follow the data would destroy exactly the signal the freeze is meant to preserve. Both values are written into the configuration file as fixed constants (Table 2.2). NB2’s principal output is a file named `cross_submatrix_risk.csv`, containing one row per pair with all of the quantities described above. This file is the sole input NB3 requires.

2.6 Machine Learning Pipeline (NB3)

2.6.1 Dataset Construction

NB3 takes the pairs produced by NB2 and applies the modelling threshold of $MIN_N = 15$, excluding those with insufficient support and labelling them INSUFFICIENT. For the pairs that remain modellable, the binary target variable is defined as

$$\text{TARGET} = \begin{cases} 1 & n_{\text{nc}} \geq \text{MIN_NC} \\ 0 & \text{otherwise,} \end{cases} \quad (2.10)$$

where $\text{MIN_NC} = 1$. A separate categorical label used for the descriptive DSS tiering in Section 2.6.6 is assigned directly from the non-conformity rate against the DSS_HIGH and DSS_MED boundaries; it is computed independently of the binary TARGET. The modellable pairs are split into a training set and a held-out test set using stratified random sampling with a twenty percent test fraction and a fixed random seed, so that class balance is preserved in both partitions.

A further point that matters for interpreting results concerns the way MIN_N operates here: it functions exclusively as a training-set inclusion filter, never as an output filter. A pair with fewer than fifteen observations is excluded from fitting the model and from the cross-validation folds. It still receives a risk score at inference time, however, since the topological features described in Section 2.6.2 are computed on the full, unfiltered dataset. These features remain available even for pairs that never contributed to training. No contaminant–submatrix combination is ever silently dropped from the final DSS output. Pairs below the training threshold are instead explicitly labelled `INSUFFICIENT` rather than omitted.

2.6.2 Feature Engineering

Eight features were engineered for the classifier, summarised in Table 2.3. Each one was deliberately constructed to be free of leakage from the target definition: `nc_rate` and `n_nc` are explicitly excluded as inputs, since both appear directly inside Equation 2.10. Two features deserve particular attention here. The first, `submat_nc_breadth`, is the fraction of all distinct contaminants that have ever been found non-conforming within a given submatrix. The second, `contam_nc_breadth`, is the fraction of all distinct submatrices in which a given contaminant has ever been found non-conforming. Both are computed on the full, unfiltered dataset before the MIN_N filter is applied. This way, the topological structure of the entire contamination landscape informs the score given to every pair, including pairs that on their own carry very little direct evidence. Neither breadth feature is computed from the `nc_rate` or `n_nc` of the specific pair being scored. Both are aggregate portfolio statistics, and as such they carry no leakage risk.

Table 2.3: Machine learning feature set (NB3, MIN_N = 15).

Feature	Encoding	Rationale
submat_nc_breadth	Ratio, pre-filter	Fraction of contaminants ever found non-conforming in this submatrix, computed on the unfiltered dataset.
contam_nc_breadth	Ratio, pre-filter	Fraction of submatrices in which this contaminant has ever been found non-conforming; same pre-filter computation.
CONTAMINANTE_enc	Label encoding	Contaminant identity, mapped to an integer code via scikit-learn’s LabelEncoder, fitted once on the full dataset.
SOTTOMATRICE_enc	Label encoding	Submatrix identity, encoded the same way.
n_total	Raw integer	Total analytical support for the pair.
log_n_total	$\log(x + 1)$	Log-scaled support, reducing the right skew of n_{total} .
seasonal_strength	Ratio	STL seasonal amplitude inherited from NB1/NB2, computed without reference to future data.
log_pair_freq	$\log(x + 1)$	Log frequency of the contaminant–submatrix pair within the overall portfolio.

Label encoding was chosen over one-hot encoding for the two categorical identity features because the submatrix variable has high cardinality. One-hot encoding handles this poorly, since it inflates dimensionality and produces sparse, uninformative columns for rare submatrices.

2.6.3 Class Imbalance and Hyperparameter Optimisation

Each of the four candidate algorithms was evaluated with its final fixed hyperparameter configuration. This configuration was selected based on preliminary experimentation and is reported in Table 2.4. Class imbalance was handled through algorithm-native weighting in each case, with no synthetic oversampling applied. The four models were then evaluated using five-fold stratified cross-validation, taking average precision (PR-AUC) as the scoring metric.

Table 2.4: Final hyperparameter configurations used for each candidate model.

Model	Parameter	Value
RandomForest	n_estimators	300
	max_depth	6
	min_samples_leaf	5
	max_features	sqrt
	class_weight	balanced
XGBoost	n_estimators	500
	max_depth	3
	learning_rate	0.04
	subsample	0.7
	colsample_bytree	0.7
	min_child_weight	10
	reg_alpha	0.5
	reg_lambda	2.0
LightGBM	n_estimators	300
	max_depth	4
	learning_rate	0.05
	num_leaves	15
	min_child_samples	20
CatBoost	iterations	300
	depth	4
	learning_rate	0.05
	l2_leaf_reg	3.0

The sensitivity of results to the choice of MIN_N and MIN_NC was evaluated by running all four models across a grid of $\text{MIN_N} \in \{5, 10, 15\}$ crossed with $\text{MIN_NC} \in \{1, 2, 3\}$ and comparing cross-validated PR-AUC across the nine resulting configurations; the outcome of that scan is reported in the results chapter. Each model used the same fixed hyperparameter configuration throughout this scan.

2.6.4 Model Evaluation

Beyond average precision, complementary metrics are reported for the final model on the held-out test set:

$$\text{Precision} = \frac{TP}{TP + FP}, \quad \text{Recall} = \frac{TP}{TP + FN}, \quad F_1 = 2 \cdot \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}}. \quad (2.11)$$

Before any model was evaluated, an overfitting criterion was fixed in advance, requiring that the absolute gap between cross-validated and held-out test PR-AUC,

$\Delta_{\text{CV-test}} = |\text{CV PR-AUC} - \text{Test PR-AUC}|$, remain below 0.05 for the chosen model to be judged acceptably generalisable. Lift over a trivial dummy classifier is reported as well, defined as cross-validated PR-AUC divided by the positive class prevalence.

2.6.5 SHapley Additive exPlanations (SHAP) Interpretability

To understand which features drive the model’s predictions and in which direction, SHapley Additive exPlanations (SHAP) values (Lundberg and Lee, 2017) are computed using TreeExplainer with interventional feature perturbation. This method marginalises over the empirical distribution of each feature rather than conditioning on the paths actually taken through the trees. The underlying decomposition is

$$\phi_i(f) = \sum_{S \subseteq N \setminus \{i\}} \frac{|S|! (|N| - |S| - 1)!}{|N|!} [f(S \cup \{i\}) - f(S)], \quad (2.12)$$

which guarantees that the contributions of all features sum exactly to the difference between a given prediction and the model’s average prediction, leaving no unexplained residual.

2.6.6 Dual-Evidence DSS Scoring Rule

The tier ultimately assigned to each pair rests on an explicit dual-evidence rule, one that draws on two independently derived sources, namely the pair’s observed non-conformity rate, measured against DSS_HIGH and DSS_MED, and the CatBoost risk probability, measured against the conventional binary classification cutoff of 0.50. Tier HIGH is assigned when the non-conformity rate reaches at least DSS_HIGH and the model’s predicted probability reaches at least 0.50. Tier MEDIUM follows when either the non-conformity rate reaches at least DSS_MED or the predicted probability reaches at least 0.50, so long as the pair has not already qualified as HIGH. Where neither condition holds, the pair is assigned tier LOW. A pair falls into INSUFFICIENT when its total analytical support drops below MIN_N. The asymmetry here is deliberate, in that the HIGH condition is conjunctive while the MEDIUM condition is disjunctive, because a false negative at the highest tier carries far graver operational consequences than a false positive does. Demanding agreement between both lines of evidence before the most consequential tier can be assigned is, in effect, a direct and explicit way of tilting the system against that costlier error.

2.6.7 Composite Score and Percentile Ranking (NB3b)

The categorical tiering is necessarily coarse. Every pair lands in one of four discrete buckets, and there is no way to compare two pairs that happen to share a tier. NB3b addresses this by computing two continuous ranking schemes for internal prioritisation within a tier. The first is a simple percentile rank of the model’s predicted probability. The second, the composite score, blends the non-conformity rate and the model probability into a single weighted figure,

$$\text{nc_rate}_{\text{norm}} = \min\left(\frac{\text{nc_rate}}{\text{DSS_HIGH}}, 1\right), \quad (2.13)$$

$$\text{composite} = w_{\text{nc}} \text{nc_rate}_{\text{norm}} + w_{\text{prob}} \text{risk_proba}, \quad (2.14)$$

with $w_{\text{nc}} = 0.45$ and $w_{\text{prob}} = 0.55$. The normalisation is anchored to DSS_HIGH rather than to a min-max rescaling over the full distribution, preventing a small number of extreme outlier rates from compressing the resolution of the score for all other pairs. A composite score of 0.70 or above is classified as HIGH and 0.35 or above as MEDIUM for descriptive comparison purposes only.

2.7 One Health Biological Analysis (NB4)

NB4 is implemented entirely in R and takes as its starting point all contaminants assigned a HIGH or MEDIUM tier, querying each against the Comparative Toxicogenomics Database (CTD) (Davis et al., 2023) to retrieve documented chemical–gene interactions. Resolving a free-text contaminant name against CTD’s naming conventions is not straightforward, so a three-level resolution strategy is applied in order, starting with an exact string match, followed by a partial match against a hand-curated dictionary using a longest-key-wins rule, and finally a regular-expression-based sanitisation step as a last resort. Each contaminant is queried twice against CTD’s curated gene interaction report. One query is restricted to *Homo sapiens* (NCBI taxon 9606), the other to *Mus musculus* (NCBI taxon 10090), and the two resulting gene sets are pooled per contaminant. A half-second delay is enforced between consecutive API calls in compliance with CTD’s terms of use. A contaminant is retained only if its pooled gene set reaches a minimum size of five.

Pathway enrichment analysis is carried out with the gseapy interface to the Enrichr API (Kuleshov et al., 2016). The KEGG 2021 Human gene set library (Kanehisa and Goto, 2000) is queried, and significance is assessed through the hypergeometric test,

$$P = \frac{\binom{K}{k} \binom{N-K}{n-k}}{\binom{N}{n}}, \quad (2.15)$$

followed by Benjamini–Hochberg correction at $\alpha = 0.05$. The contaminant, gene, and pathway sets are assembled into a single tripartite directed graph using `igraph`, in which an edge runs from a contaminant node to a gene node whenever CTD documents an interaction, and from a gene node to a pathway node whenever that gene is a member of that pathway according to the KEGG enrichment result. A gene node is classified as a hub gene if its contaminant degree, the number of distinct contaminants linked to it, is two or greater. The network is visualised with `ggraph` using the Kamada–Kawai algorithm and exported in GraphML, PNG, and tabular form.

2.8 Prospective and Retrospective Blind Validation

The trained CatBoost model, together with the DSS thresholds fixed in Section 2.5, was applied without retraining and without recalibration to two independent hold-out datasets extracted from the LIMS. One is prospective, covering January through May 2026. The other is retrospective, covering the whole of calendar year 2023. Testing in both temporal directions, rather than in the prospective direction alone, gives a stronger indication of whether the eight features capture something durable about contamination risk. Good performance on a year that precedes the training data entirely cannot be attributed to mere proximity in time to the training distribution.

The two evaluation windows differ considerably in length, and this difference motivated deriving evaluation parameters separately for each one. In the five-month 2026 window, pairs with exactly one recorded non-conformity event turn out to be concentrated overwhelmingly among pairs with very little analytical support. That pattern is consistent with sampling noise. The evaluation subset therefore applies $\text{MIN_N} = 5$ and $\text{MIN_NC} = 2$. In the twelve-month 2023 window, by comparison, pairs with exactly one recorded non-conformity event show substantially higher mean support, which points to a plausibly genuine low-level signal. Here the evaluation subset retains $\text{MIN_N} = 15$ but relaxes the non-conformity requirement to $\text{MIN_NC} = 1$. The DSS tier thresholds, DSS_HIGH and DSS_MED , are not recalculated on either blind window. Both evaluations apply the same fixed values, derived once from the training distribution.

The acceptance criteria differed by direction rather than following a single shared

threshold. A 2026 result was accepted when its PR-AUC gap relative to the training cross-validated value stayed below 0.10 and its ROC-AUC gap did the same. Looking backward to 2023, the ROC-AUC criterion held at that same level, but the PR-AUC threshold was loosened to below 0.15. This reflects the lower positive rate expected in a twelve-month window whose analytical composition differs from that of the training period. In both directions, the proportion of previously unseen contaminant–submatrix combinations is reported as a supplementary diagnostic.

2.9 Deployment and Reproducibility

The DSS output reaches users through three separate interfaces, each tailored to a distinct community. Operators who need the most current view of the data are served by a Streamlit dashboard, refreshed live through a direct query connection to the LIMS. Streamlit itself is a Python framework for building interactive data applications directly from a script, with no separate front-end code required. It was chosen here because the dashboard could be written entirely in Python, reusing the same data-processing logic as the rest of the pipeline. For users who need an offline, portable snapshot without LIMS access, a standalone, self-contained HTML report is produced by a purpose-built export script (`export_simple.py` and a compact variant `export_simple_compact.py`), with all data embedded directly into the page as JSON at generation time. This export includes an interactive table, trend visualisations, and a client-side Excel download function implemented entirely within the HTML file. Non-technical stakeholders already accustomed to Power BI are served by a dedicated dashboard, connected via DirectQuery to the same SQL Server instance. All three interfaces consume the same frozen output of NB3b and NB4 and apply no further transformations. The risk scores displayed are therefore identical no matter which interface is used to access them. Reproducibility of the analytical results rests on the combination of Docker containerisation, the single centralised YAML configuration, the fixed random seed, and the structured, timestamped archiving of every executed notebook and every intermediate output, in line with the FAIR principles for reproducible computational workflows (Wilkinson et al., 2016).

Chapter 3

Results and Discussion

3.1 Dataset Overview and Distributional Properties

Following the cleaning and aggregation steps described in NB1 (Section 2.3), the surveillance dataset comprised 182,033 certified analytical results spanning 122 distinct contaminants and 640 food submatrices, covering the full twenty-four-month window from January 2024 to December 2025. Across the entire dataset, the overall non-conformity rate was 3.3%. How this rate varies across sections is shown in Figure 3.1. Table 3.1 summarises the key statistics for each section, including volume, the most critical contaminant, and any significant temporal trends.

Table 3.1: Section-level summary for the training window (January 2024–December 2025). NC rate is the fraction of valid results exceeding the applicable maximum level. MK trends are contaminants with a significant Mann–Kendall result on the STL-decomposed monthly series.

Section	N valid	Cont.	NC%	Most critical		MK trends
S1 Mycotoxins	102,768	67	3.9	Fumonisin		DON, Fumonisin, HT2, ZEA
S2 Plant toxins	1,374	20	16.1	Tropane	alk.	Opiates
S3 Heavy metals	52,412	8	2.8	Mercury (26.5%)		none
S4 POPs/PFAS	8,257	12	3.0	Dioxin-PCBs		none
S5 Process	12,751	8	0.7	3-MCPD (10.8%)		none
S6 Nitrates	4,471	7	0.3	Nitrates (3.3%)		none
Total	182,033	122	3.3			

Taken together, Figure 3.1 and Table 3.1 show that analytical volume and non-conformity risk run in almost opposite directions across sections. Section 1, Mycotoxins, accounts for the great majority of analytical volume yet sits at a comparatively modest 3.9% non-conformity rate, below the 5% alert threshold; this

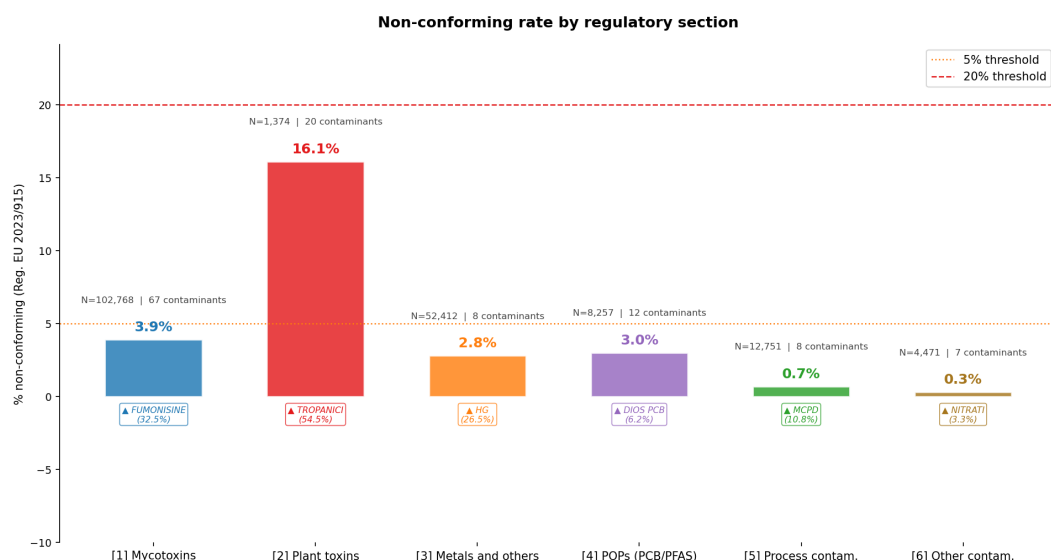


Figure 3.1: Non-conformity rate by regulatory section, with reference lines at the 5% alert threshold and the 20% critical threshold.

section-wide figure conceals a much higher rate for the summed Fumonisin B1+B2 parameter specifically, at 32.5%, consistent with the well-documented tendency of maize-derived matrices to be co-contaminated by several *Fusarium*-derived mycotoxins simultaneously. Section 2, Plant toxins, is the clear outlier in the opposite direction: with only 1,374 analytical results, less than one percent of total volume, it still records an overall non-conformity rate of 16.1%, the only section to cross the 5% alert threshold, driven overwhelmingly by the Tropane alkaloid group at 54.5%. The remaining four sections, Heavy metals, POPs/PCBs/PFAS, Process contaminants, and Nitrates/perchlorates, all sit comfortably below threshold at 2.8%, 3.0%, 0.7%, and 0.3% respectively. This section-to-section heterogeneity in baseline risk is precisely what motivates testing for enrichment separately within each section rather than pooling the whole dataset together.

3.2 Temporal Dynamics of Non-Conformity Rates

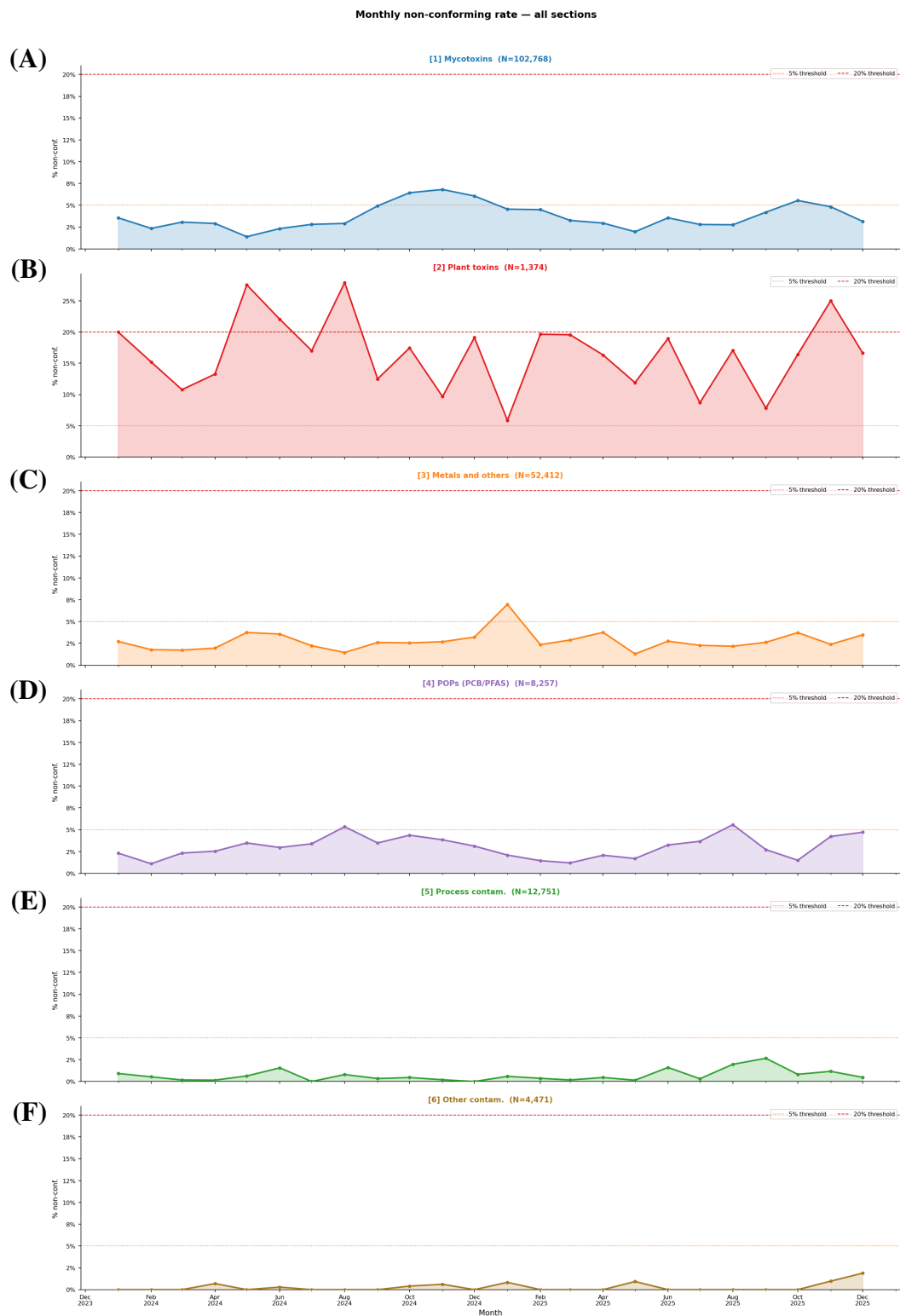


Figure 3.2: Monthly non-conformity trends across all six regulatory sections, January 2024–December 2025, with reference lines at 5% and 20%: (A) mycotoxins; (B) plant toxins; (C) metals and other contaminants; (D) persistent organic pollutants (PCBs and PFAS); (E) process contaminants; and (F) other contaminants.

Applying the seasonal-trend decomposition and Mann–Kendall procedure described in the previous chapter to each section’s monthly series reveals two qualitatively distinct temporal regimes, visible in Figure 3.2. Panel A, regarding mycotoxins, shows

a clearly seasonal pattern, with non-conformity rates rising during specific months and falling back outside that window. In Italy, maize is harvested in September–October, while wheat and barley are harvested earlier, between May and July; the observed peak falls within the maize harvest window, so it cannot be attributed to wheat or barley on this basis. Once the seasonal component is separated out via STL, the underlying secular trend over the full two years is not statistically significant. This timing coincidence does not, on its own, establish a harvest-related origin for the peak, since the LIMS records a sample’s arrival date at the laboratory, not the harvest date, so the interval between the two cannot be established for individual samples. Panel B, which displays data about plant toxins, shows a different pattern, with essentially no resolvable seasonal structure: its elevated non-conformity rate is sustained at a roughly constant, persistently high level across almost the entire observation window rather than oscillating with the agricultural calendar, a pattern far more consistent with a structural, supply-chain-level problem than with a recurring seasonal effect. The distinction carries operational weight, since a seasonal pattern can be anticipated and planned around, whereas a persistent structural elevation calls for a different kind of intervention. The remaining four sections (Panels C–F) show comparatively stable, low-amplitude behaviour throughout the two years, with no section-level non-conformity rate exceeding the 5% threshold in any individual month, lending confidence to using their historical rates directly as inputs to the DSS without correcting for strong temporal drift.

3.3 Statistical Enrichment Analysis

Section-stratified Fisher exact testing, corrected for multiple comparisons within each section using the Benjamini–Hochberg procedure described in Section 2.5, identified 75 significantly enriched contaminant–submatrix pairs. These were drawn from 3,496 pairs tested at the exploratory stage, one more than the 3,495 pairs carried forward to the modelling and DSS stages. The difference arises because the enrichment analysis admits a pair on the looser support threshold of $\text{MIN_N} = 5$, whereas the modelling stage applies the stricter $\text{MIN_N} = 15$, so a single pair clears the first bar but not the second. Against the 3,496 pairs tested, the 75 enriched pairs amount to a rate of 2.1%. Table 3.2 breaks this result down by section.

Table 3.2: Enrichment results by regulatory section ($q < 0.05$, Benjamini–Hochberg corrected). Top pairs are discussed in the text.

Section	Enriched	Tested	Rate	Max OR	Min q
S1 Mycotoxins	25	1,573	1.6%	23.15	3.93×10^{-119}
S2 Plant tox.	2	87	2.3%	4.42	3.73×10^{-9}
S3 Heavy met.	38	1,153	3.3%	90.05	1.90×10^{-144}
S4 POPs/PFAS	6	290	2.1%	392.76	1.24×10^{-22}
S5 Process	4	200	2.0%	7.86	1.67×10^{-6}
S6 Nitrates	0	193	0.0%	n/a	n/a
Total	75	3,496	2.1%		

Across the whole dataset, no odds ratio exceeds that of the summed PCB congener parameter in fresh fish, found in Section 4 at 392.76, fitting the well-established tendency of polychlorinated biphenyls to biomagnify progressively through aquatic food chains given their high lipophilicity and environmental persistence. The largest number of enriched pairs falls within Section 3, Heavy metals, where 38 of 1,153 tested pairs survive correction; cadmium in fresh fish leads this group at an odds ratio of 90.05, in line with its well-documented bioaccumulation in marine organisms. Section 1 contributes the bulk of the dataset’s analytical volume, yet its enrichment rate is comparatively modest at 1.6%. Aflatoxin B1 in maize germ stands as the most strongly enriched pair by q -value, at 3.93×10^{-119} , though its odds ratio of 23.15 is more moderate than several heavy metal pairs: the large sample size for this combination drives extreme statistical significance even without an extreme effect size.

Section 2 illustrates a more subtle methodological point that follows directly from how the Fisher test’s background is constructed here: each contaminant–submatrix cell is tested against the same contaminant’s own non-conformity rate across its other submatrices within the section, not against the section as a whole. The Tropane alkaloid group records the highest contaminant-level non-conformity rate in Section 2 at 54.5%, and individual Scopolamine pairs record rates at or near 100% in specific submatrices, yet these contaminants do not register among the significantly enriched pairs. This is not a contradiction: if Scopolamine’s non-conformity rate is elevated fairly uniformly across most submatrices it is tested in, rather than concentrated disproportionately in one relative to its others, no single submatrix stands out as anomalous for that contaminant, even though its absolute risk level is exceptionally high.

The practical lesson is that this enrichment test answers a narrower question than it might first appear: whether a given submatrix is unusually problematic for a contaminant that otherwise behaves normally, not whether the contaminant’s absolute risk level is itself high. The two questions need to be read together rather

than as substitutes, which is precisely why the dual-evidence DSS scoring rule introduced in Section 2.6.6 deliberately combines the enrichment result with an entirely independent source of evidence, the model’s own predicted risk probability, and why the descriptive overview in Section 3.1 reports Scopolamine’s absolute non-conformity rate directly rather than relying solely on enrichment to flag it.

Section 6, Nitrates and perchlorates, registers zero enriched pairs, consistent with its very low overall non-conformity rate and with the largely agronomic, rather than supply-chain, determinism of nitrate accumulation in crops, where variation within the contaminant class tends to swamp any systematic difference between submatrices.

3.4 Machine Learning Model Comparison and Selection

3.4.1 Sensitivity to MIN_N and MIN_NC

Before any model was finalised, all nine combinations of $\text{MIN_N} \in \{5, 10, 15\}$ and $\text{MIN_NC} \in \{1, 2, 3\}$ described in the previous chapter were evaluated, and the best-performing model obtained at each configuration was compared on cross-validated PR-AUC. The configuration ultimately adopted paired $\text{MIN_N} = 15$ with $\text{MIN_NC} = 1$. It offered the most favourable balance between the size of the resulting training set, 1,518 pairs, and the reliability of the labels attached to those pairs. A stricter threshold of $\text{MIN_N} = 20$ was also examined. It shrinks the pool of eligible pairs still further, and this loss was not compensated by any meaningful gain in cross-validated performance. For that reason it was not adopted as the final configuration.

3.4.2 Model Comparison

Table 3.3: Model comparison at $\text{MIN_N} = 15$, $\text{MIN_NC} = 1$ (1,518 pairs, 286 positives, 18.8%), following the hyperparameter configuration selected through preliminary experimentation (Section 2.6.3). F_1 is reported for the positive class (Risk) on the held-out test set.

Model	CV PR-AUC ($\pm$ SD)	Test ROC-AUC	Test PR-AUC	F_1 (Risk)
CatBoost (selected)	0.8360 ($\pm$0.0267)	0.9504	0.8107	0.72
XGBoost	0.8288 ($\pm$ 0.0298)	n/a	n/a	n/a
Random Forest	0.8121 ($\pm$ 0.0410)	n/a	n/a	n/a
LightGBM	0.8060 ($\pm$ 0.0222)	n/a	n/a	n/a

Table 3.3 reports the comparative performance of all four models. CatBoost was selected as the final model, achieving the highest cross-validated PR-AUC at 0.8360, ahead of XGBoost at 0.8288, Random Forest at 0.8121, and LightGBM at 0.8060. Its edge can be traced structurally to its native handling of high-cardinality categorical features through an internal ordered target encoding scheme, which sidesteps the leakage risk a pre-computed target encoding applied to the full dataset would otherwise introduce. Between the cross-validated PR-AUC of 0.8360 and the held-out test PR-AUC of 0.8107 sits a gap of only 0.0254, comfortably inside the pre-specified overfitting criterion of less than 0.05, suggesting that the leakage-free feature design performs as intended. The lift over a dummy classifier, 4.43 times the baseline PR-AUC, confirms the model is extracting substantial real discriminative signal rather than simply reproducing the class prior. The test ROC-AUC of 0.9504 points to strong overall rank-ordering capacity, but the PR-AUC of 0.8107 is the more operationally meaningful figure here: with positive pairs making up only 18.8% of the training set, ROC-AUC can remain misleadingly high even for a mediocre classifier, since it is rewarded heavily for correctly rejecting the large majority class, whereas PR-AUC is calculated with reference to the positive class specifically and is far less forgiving of a model that performs well only on the easy majority of cases.

3.4.3 Calibration and Error Structure

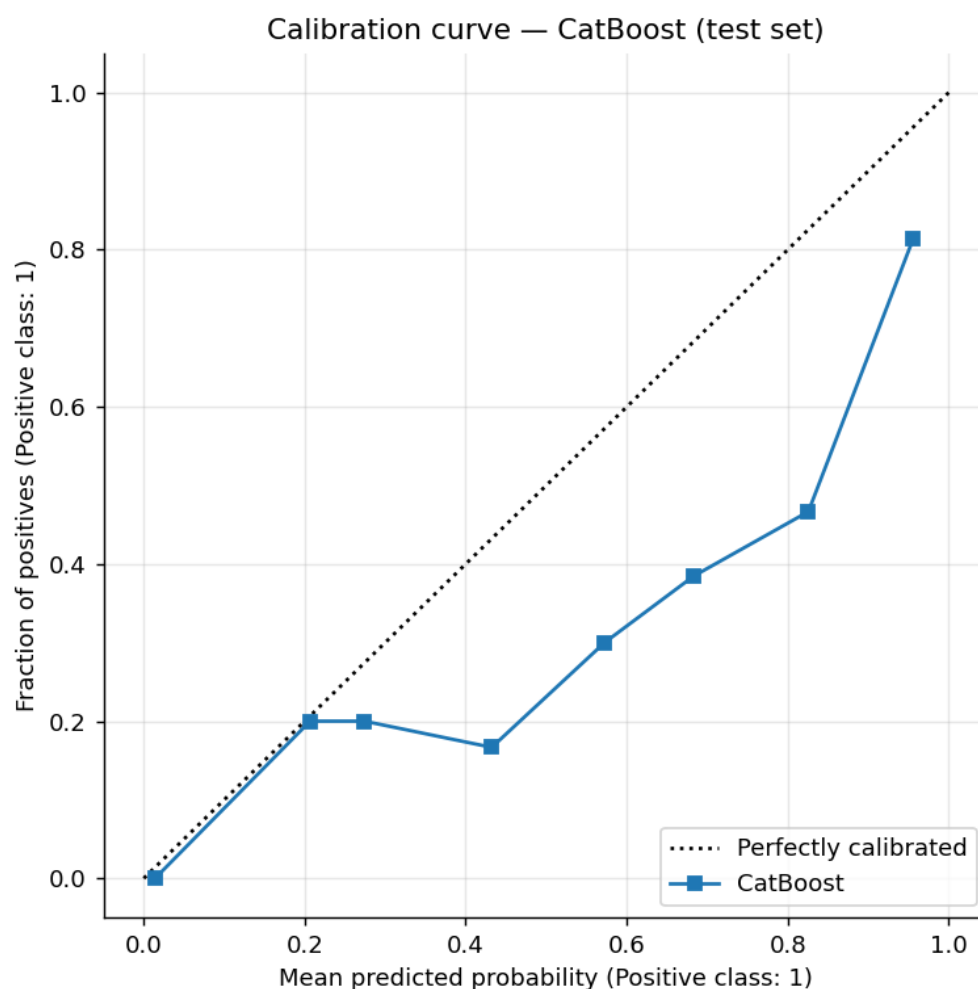


Figure 3.3: Calibration curve on the held-out test set: predicted probability against observed fraction of positives, in eight bins.

Figure 3.3 shows how closely the model’s predicted probabilities track the actual observed risk. Calibration stays close to ideal across most of the range, with a slight tendency towards overconfidence between 0.4 and 0.6; above roughly 0.6, the model’s output can be read as a reasonably well-calibrated absolute probability rather than merely a relative ranking. At the standard 0.50 classification threshold, the held-out test set yields 50 true positives, 7 false negatives, 31 false positives, and 216 true negatives, out of 304 pairs in total. These false positives are not spread randomly: they cluster among pairs with a high predicted risk probability paired with an observed non-conformity rate at or near zero. That pattern does not necessarily point to a flawed model. The two dominant features driving the model, described in the next section, are computed from the breadth of non-conformity across the entire contamination landscape rather than from the direct history of the individual pair in question, so a pair can legitimately score high on the strength of its submatrix’s or contaminant’s broader pattern even without a non-conforming result of its own yet.

Whether such a case is a genuine false alarm or an early, still-unconfirmed signal is a distinction the available data cannot yet resolve; an operator using this system should keep that in mind rather than treat every high-scoring, zero-history pair as either a confirmed risk or a model error.

3.5 SHAP Explainability Analysis

Table 3.4: Global SHAP feature importance (CatBoost, MIN_N = 15, $n_{\text{pairs}} = 1,518$, $n_{\text{pos}} = 286$).

Rank	Feature	Mean $ \phi_i $ and interpretation
1	submat_nc_breadth	2.509, fraction of contaminants exceeding their limit in this submatrix, computed before the MIN_N filter.
2	contam_nc_breadth	2.001, fraction of submatrices in which this contaminant exceeds its limit, same pre-filter computation.
3	CONTAMINANTE_enc	0.351, contaminant identity, label-encoded.
4	log_n_total	0.233, log-scaled analytical support.
5	n_total	0.194, raw sample count, retained for tree interaction effects.
6	SOTTOMATRICE_enc	0.176, submatrix identity, label-encoded, 640 categories.
7	seasonal_strength	0.116, STL seasonal component strength.
8	log_pair_freq	0.000, log pair frequency; negligible independent contribution.

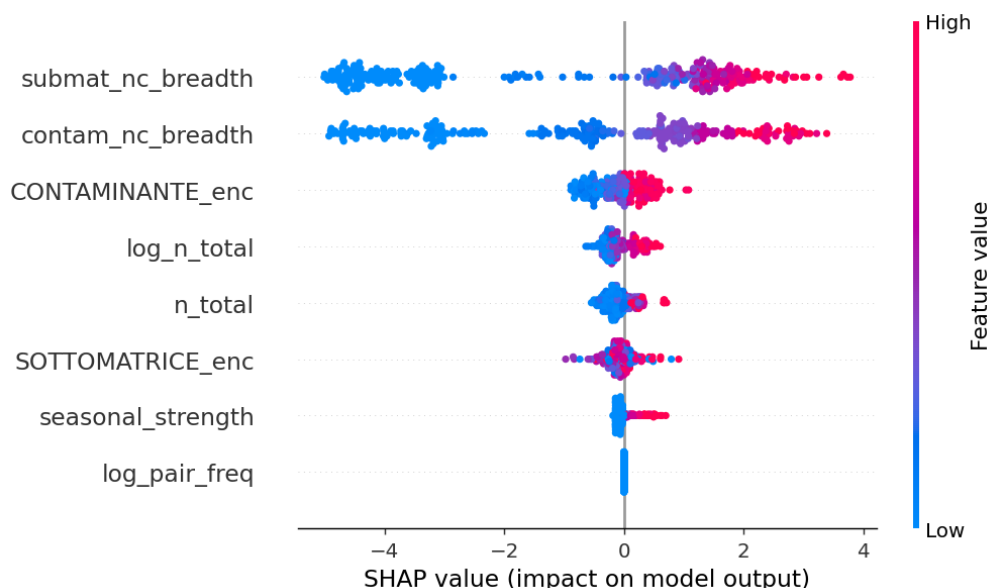


Figure 3.4: SHAP summary plot for the held-out test set, CatBoost, all eight features ranked by mean absolute SHAP value. Point colour encodes each feature’s value for that observation, allowing the direction of each feature’s effect to be read alongside its magnitude.

Table 3.4 lists the global SHAP feature importance values, confirming quantitatively what the calibration discussion above already suggested qualitatively. Figure 3.4 makes the same pattern visible at the level of individual predictions, not merely as an aggregate ranking. The two breadth features dominate the model overwhelmingly, together contributing a combined mean absolute SHAP value of 4.510 against only 1.069 for all six remaining features combined. In practical terms, the single strongest determinant of a pair’s predicted risk is not the specific identity of the contaminant or submatrix, nor even the direct sampling history of that exact pair, but the broader topology of where else, across the entire 3,495-pair landscape, non-conformity tends to cluster.

The point colouring in Figure 3.4 confirms this relationship runs in the expected direction throughout the test set: higher values of `submat_nc_breadth` and `contam_nc_breadth` are consistently associated with a positive SHAP contribution, pushing predicted risk upward, and the relationship does not reverse for any subset of pairs. A submatrix that already hosts several other problematic contaminants, or a contaminant that already causes problems across several other submatrices, pushes the predicted risk of a given pair upward even when that specific pair’s own recorded history is thin or entirely clean.

This is the intended behaviour of a topological feature design: it lets the model extend meaningful risk estimates to pairs with too little direct history to support a confident estimate on their own, by borrowing strength from the wider pattern they sit within. It is also why the model’s high-confidence errors take the specific shape discussed above, since a pair can score highly on structural grounds alone without yet having

accumulated direct confirming evidence.

The last feature, `log_pair_freq`, contributes essentially nothing once the other seven are accounted for: how often a contaminant–submatrix combination appears in the dataset carries no useful signal of its own once its breadth, identity, and support are already known.

3.6 Composite DSS Risk Ranking

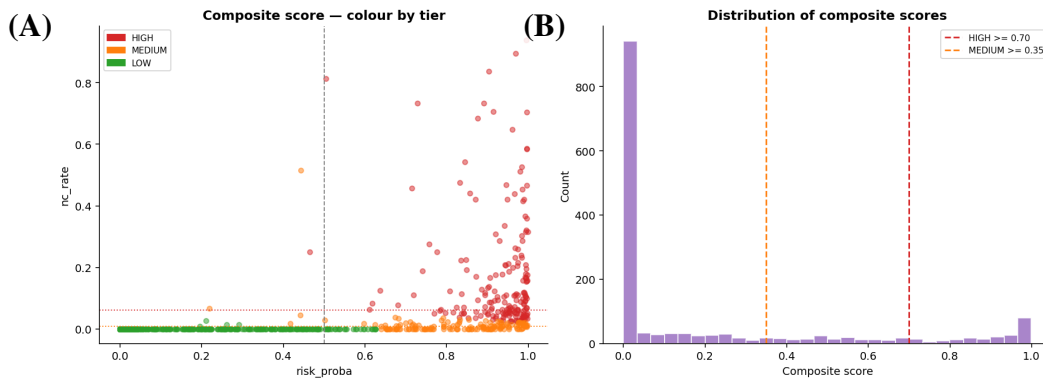


Figure 3.5: Composite DSS risk ranking across all 3,495 pairs. (A) Observed non-conformity rate against CatBoost risk probability, coloured by DSS tier. (B) Distribution of the composite score, with the MEDIUM and HIGH tier thresholds shown as dashed lines.

Applying the dual-evidence scoring rule defined in Section 2.6.6 to the full set of 3,495 contaminant–submatrix pairs produces the tier distribution summarised in Table 3.5. Panel A of Figure 3.5 shows the pair-level relationship between the two evidence sources, whereas Panel B shows the continuous composite score distribution that underlies this discrete tiering.

Table 3.5: DSS tier distribution across all 3,495 contaminant–submatrix pairs.

Tier	Pairs	% of total
HIGH (both conditions satisfied)	127	3.6%
MEDIUM (one condition satisfied)	248	7.1%
LOW (neither condition satisfied)	1,143	32.7%
INSUFFICIENT ($n_{\text{total}} < 15$)	1,977	56.6%
Total	3,495	100.0%



Figure 3.6: Top 40 HIGH-tier contaminant–submatrix pairs ranked by risk_proba, coloured by observed non-conformity rate.

Just over half of all pairs, 56.6%, fall into the INSUFFICIENT category. This follows directly from the long tail inherent in a combination space of 640 submatrices crossed with 122 contaminants, where the great majority are tested only sporadically together. As emphasised in the previous chapter, this designation governs eligibility for model training only. Every one of these 1,977 pairs still receives a full risk score at inference time through the breadth features, which remain available regardless of

how thin that specific pair's own history is. Figure 3.6 makes visible a pattern that recurs throughout the HIGH-tier results. Three largely distinct mechanistic clusters dominate the top of the ranking.

The most prominent is multi-mycotoxin co-contamination of maize germ. It recurs across seven separate entries in the top 40, spanning aflatoxin B1, zearalenone, the summed fumonisins, HT-2, DON, and their combined sum. This pattern converges with, rather than merely repeats, the structural breadth signal already identified as the dominant SHAP feature. Maize germ's elevated risk is visible both in its direct enrichment statistics and in the topological feature that captures how many distinct contaminants tend to cluster within it.

The second cluster is pyrrolizidine alkaloid contamination across botanical matrices. It appears five times, for fresh borage, organic chamomile, decaffeinated tea leaves, chamomile flower, and cumin seed, which is consistent with the genotoxic, no-safe-threshold status of this contaminant class discussed in Chapter 1. The third cluster, smaller still, is lipophilic bioaccumulation in fresh fish, where mercury and the summed PCB congeners both appear for the same matrix. This too is consistent with the extreme odds ratio already noted in the enrichment analysis.

3.7 Bidirectional Temporal Validation

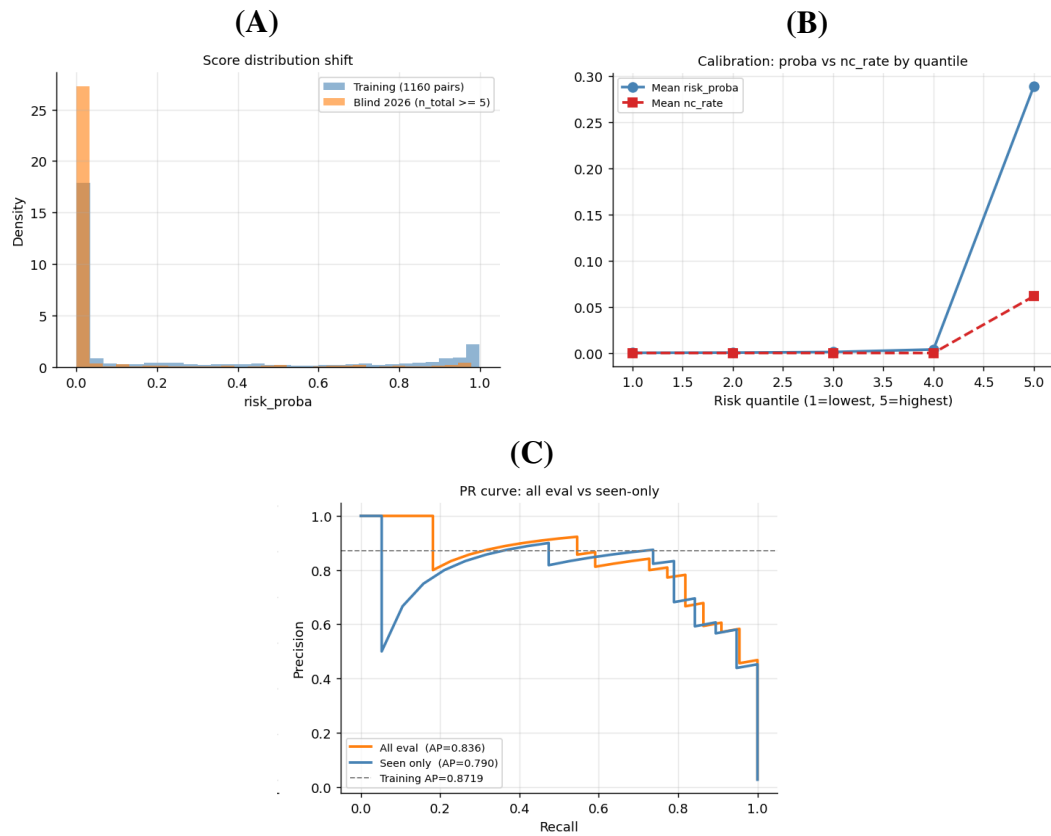


Figure 3.7: Prospective blind validation (January–May 2026; 751 pairs): (A) predicted risk distribution relative to training; (B) predicted risk and observed non-conformity rate by quantile; and (C) precision–recall curves for all and previously seen pairs.

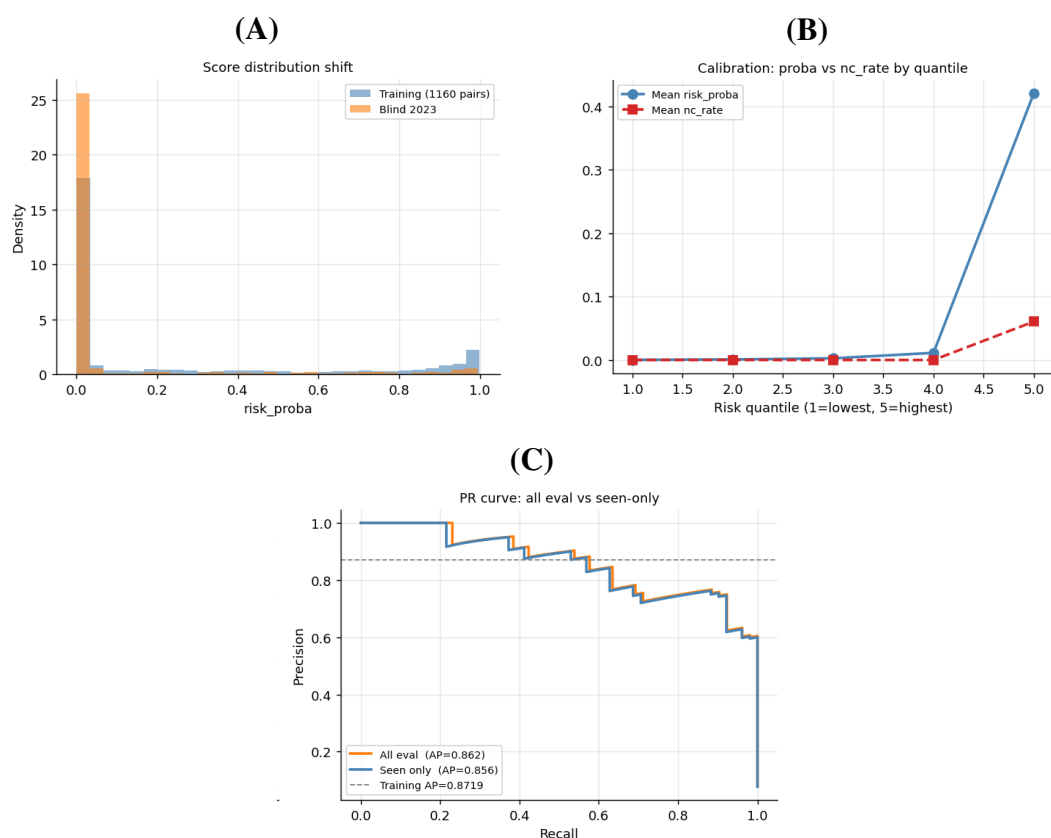


Figure 3.8: Retrospective blind validation (January–December 2023; 668 pairs): (A) predicted risk distribution relative to training; (B) predicted risk and observed non-conformity rate by quantile; and (C) precision–recall curves for all and previously seen pairs.

Table 3.6: Bidirectional temporal validation summary. Gaps relative to training CV PR-AUC (0.8360) and test ROC-AUC (0.9504); negative gap = exceeds training reference.

Metric	Training	2026 Prosp. (<i>n</i> = 751)	2023 Retro. (<i>n</i> = 668)	Verdict
PR-AUC	0.8360	0.8362	0.8622	both PASS
ROC-AUC	0.9504	0.9944	0.9881	both PASS
PR-AUC gap	n/a	−0.0002	−0.0262	both improved
ROC-AUC gap	n/a	−0.0440	−0.0377	both improved
Gap threshold	n/a	< 0.10	< 0.15	both met
Positive rate	18.8%	2.9%	7.8%	see text
Window	24 mo	5 mo	12 mo	n/a
MIN_N (eval)	15	5	15	window-specific
MIN_NC (eval)	1	2	1	window-specific

Figures 3.7 and 3.8 present the same three diagnostics for the prospective and retrospective windows, respectively: Panel A shows the predicted probability distribution relative to training, Panel B compares mean predicted risk probability with observed non-conformity rate by risk quantile, and Panel C shows precision–recall performance for all evaluated pairs and for previously seen pairs only. Table 3.6 reports the quantitative performance metrics for both directions alongside the training

reference values. Both blind evaluations satisfy the acceptance criteria fixed before either was run, though those criteria, and the evaluation subsets they were applied to, were not identical between the two directions. As detailed in Chapter 2, the 2026 window required a less restrictive minimum support, $\text{MIN_N} = 5$, but a stricter minimum non-conformity count, $\text{MIN_NC} = 2$: pairs with exactly one recorded non-conformity event over such a short window were concentrated among pairs with very little support, and were judged sampling noise rather than signal. The 2023 window instead retained the training support threshold, $\text{MIN_N} = 15$, and the training non-conformity count, $\text{MIN_NC} = 1$; pairs with a single recorded event over a full twelve months carried, on average, well over a hundred underlying observations, and were judged a genuine low-level signal. The PR-AUC acceptance margin was relaxed from 0.10 to 0.15 for the 2023 window alone, given its lower expected positive rate, while the ROC-AUC margin remained unchanged in both directions. Despite these window-specific adjustments, both directions exceed rather than merely approach the training reference values for PR-AUC and ROC-AUC alike, which is the central empirical finding of this thesis: a model trained exclusively on the 2024–2025 window, with its decision thresholds fixed on that same window and never subsequently adjusted, retains its discriminative capacity both forward, on data generated after the window closed, and backward, on data generated before the window began. This bidirectional agreement is a considerably stronger demonstration of stability than a conventional forward-only prospective test, since it is difficult to attribute good performance on a year that precedes the training data entirely to temporal proximity or a fortuitously stable data distribution. The most parsimonious explanation is that the eight topological features genuinely capture durable structural properties of which submatrices and contaminants are intrinsically prone to non-conformity, properties that hold across a span of years rather than being an artefact specific to the two years used for training.

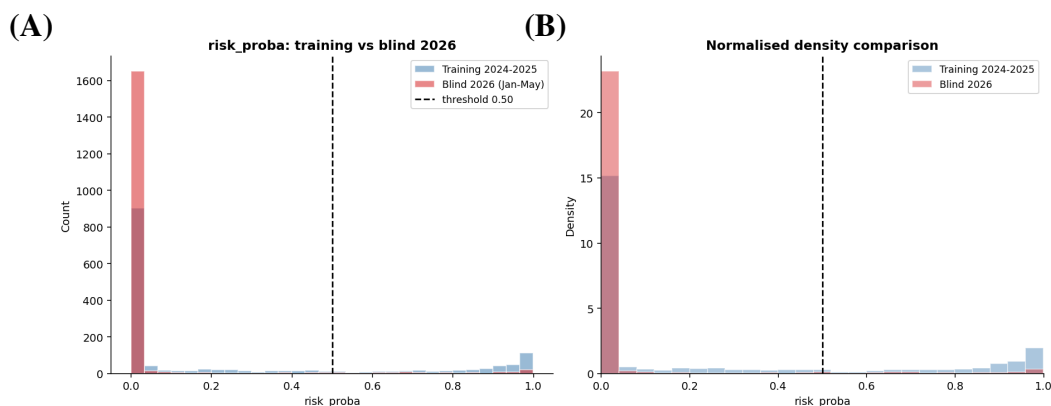


Figure 3.9: Comparison of predicted risk probabilities between the training set and the 2026 prospective validation window. (A) Overlaid count distributions. (B) Normalised density distributions.

The positive rate falls from 18.8% in the training set to 7.8% in the twelve-month 2023

window, and drops further to 2.9% in the five-month 2026 window. Figure 3.9 clarifies why this should not be read as evidence of model failure: the target definition requires at least one non-conformity event within the window (raised to two only for the shorter 2026 prospective subset), so a shorter window affords fewer opportunities to cross that bar even if the underlying per-sample probability has not changed. Normalising each positive rate by window length gives a roughly consistent ratio across all three periods, around 0.78% per month for training, 0.65% for 2023, and 0.58% for 2026, the signature of a labelling artefact rather than a genuine change in the contamination process. Panels A and B of Figure 3.9 show, in absolute and normalised terms, respectively, that the shape of the predicted probability distribution is preserved between training and the 2026 prospective window, with mass concentrated near zero for low-risk pairs and a separate cluster near one for high-risk pairs. The analogous retrospective comparison is shown in Panel A of Figure 3.8. Thus, the model has not collapsed towards a constant output under either blind window.

3.8 One Health Biological Context: Hub Genes and Shared Pathways

The 375 pairs assigned tier HIGH or MEDIUM by the DSS scoring rule correspond to 47 distinct contaminants. Each was queried against the Comparative Toxicogenomics Database, against both the human and the mouse curated interaction sets, as described in the previous chapter. This pooled query returned 405 unique gene targets, of which 261, 64.4% of the total, qualify as hub genes under the degree-two criterion, each documented as interacting with at least two of the 47 contaminants independently. The resulting tripartite network contains 27,609 edges in total. The human and mouse results are pooled rather than reported separately, so the network does not distinguish which specific interactions were established in human tissue versus murine models, a point addressed directly as a limitation in Chapter 4.

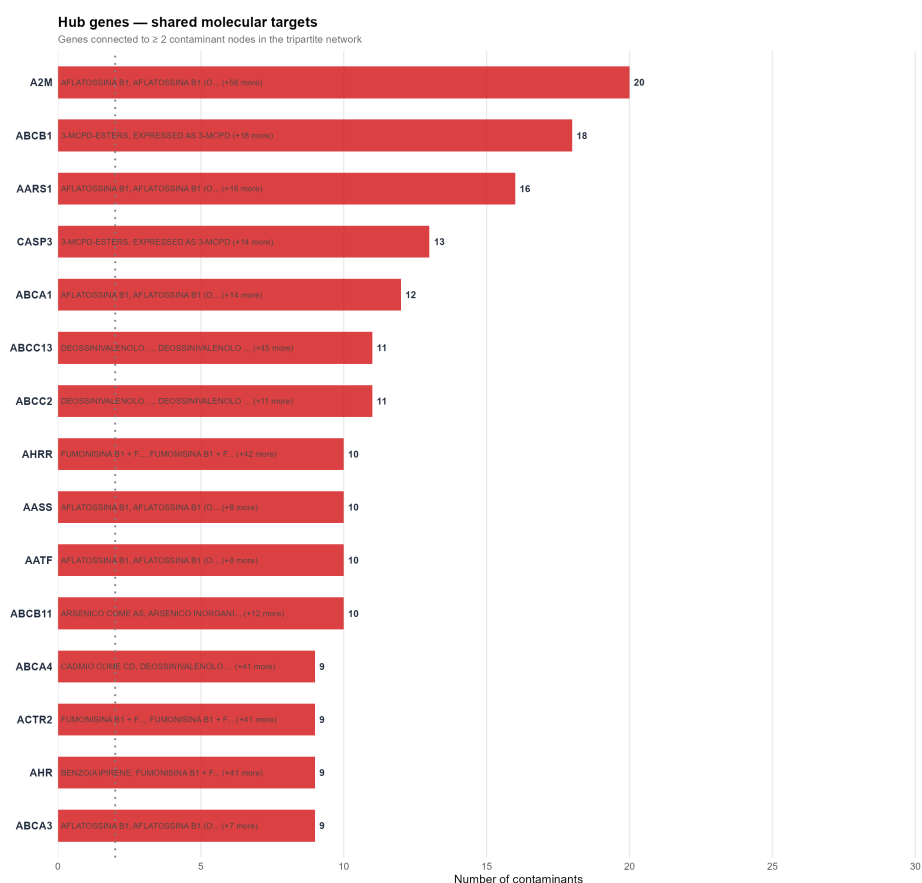


Figure 3.10: Top CTD hub genes by contaminant degree, among the 47 HIGH and MEDIUM tier contaminants, 405 CTD genes, 261 hub genes with degree at least two.

Figure 3.10 ranks these hub genes by how many distinct contaminants each one is linked to. *ABCB1*, which encodes the P-glycoprotein efflux pump, is linked to 18 of the 47 contaminants and is the second most connected. P-glycoprotein actively limits the cellular uptake and tissue distribution of a wide range of xenobiotics. If eighteen structurally unrelated dietary contaminants compete for the same finite transport capacity, simultaneous exposure to several of them at once could in principle increase the systemic bioavailability of all of them together. That is a pharmacokinetic interaction no framework assessing one contaminant at a time against its own maximum level could detect. *A2M*, linked to 20 contaminants and carrying the highest absolute contaminant degree in the network, points to convergent perturbation of acute-phase inflammatory regulation. *AARS1*, linked to 16 contaminants, implicates core translational machinery as a shared target. *CASP3*, linked to 13 contaminants, indicates convergent pro-apoptotic signalling across mycotoxins, process contaminants, and plant toxins at once. One caveat deserves to be stated plainly here: CTD documents annotated chemical-gene interactions, not measured binding affinities or transport kinetics under dietary exposure conditions. These findings are best read as identifying biologically plausible mechanisms worth further investigation, not as demonstrated interactions in vivo.

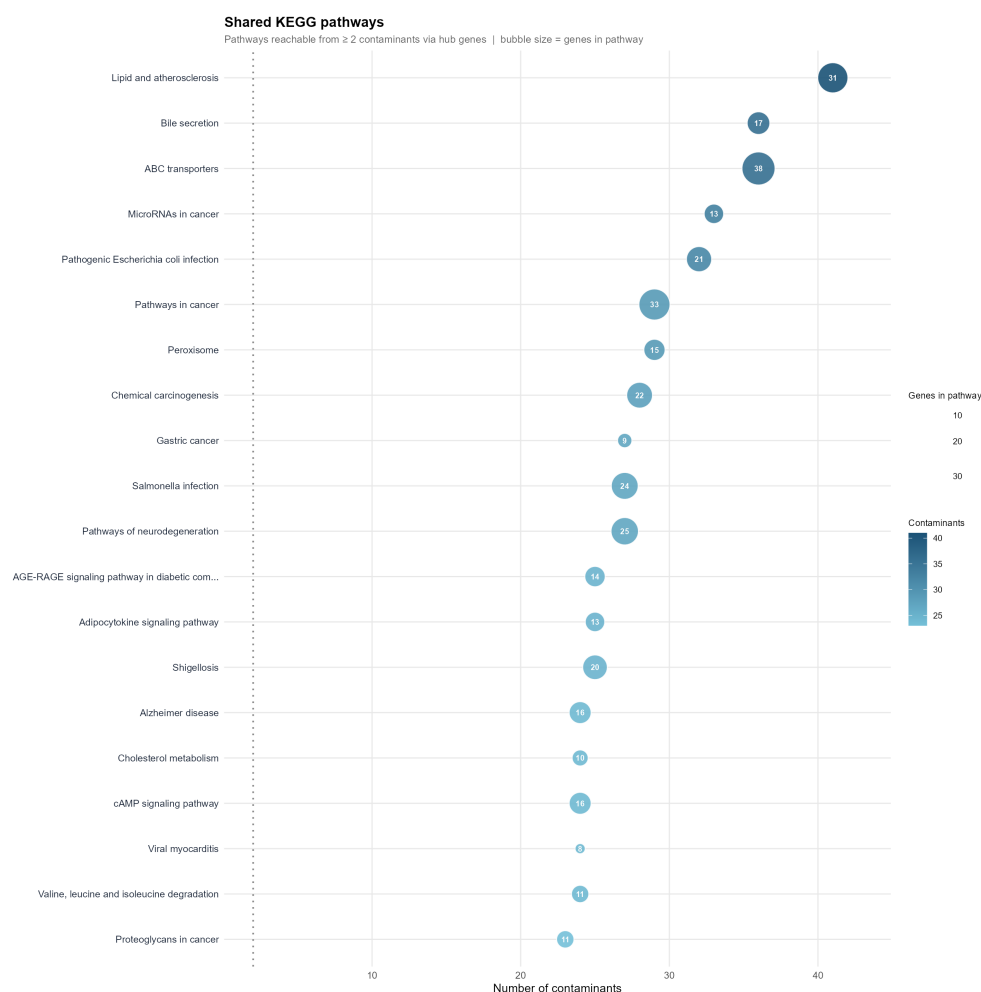


Figure 3.11: Shared KEGG pathways across the 47-contaminant portfolio. 252 of 263 enriched pathways, 95.8%, are shared between at least two contaminants via common hub genes.

The pathway-level enrichment in Figure 3.11 reinforces this picture of convergence rather than independence. “Lipid and atherosclerosis” stands as the most broadly shared pathway in the entire analysis, reached by 41 of the 47 contaminants and spanning mycotoxins, heavy metals, and persistent organic pollutants together. It offers a molecular mechanistic basis for the cardiovascular risk associations that dietary contaminant exposure has long been suspected of carrying, an association the current single-contaminant regulatory framework has no mechanism for quantifying directly. “ABC transporters”, reached by 36 contaminants, corroborates the *ABCB1* finding from the gene-level analysis at the pathway level, and it also implicates *ABCC2* and *ABCA1* in the same broad efflux-competition mechanism. “Bile secretion”, reached by 36 contaminants, points to convergent hepatic and gastrointestinal burden. “Pathways in cancer”, reached by 29 contaminants, is consistent with the IARC carcinogen classifications already attached to several members of this portfolio, among them aflatoxin B1, benzo[a]pyrene, inorganic arsenic, and the dioxin-like PCBs. Taken as a whole, the fact that 252 of 263 enriched pathways are shared between

two or more contaminants demonstrates that this 47-contaminant portfolio does not behave, biologically, as forty-seven independent risks. It converges repeatedly on a comparatively small number of shared molecular bottlenecks and shared biological processes. This convergence, entirely absent from the per-contaminant structure of Regulation (EU) 2023/915 itself, offers a mechanistic rationale for why the DSS's prioritisation of this particular set of pairs is defensible not only statistically and predictively but biologically as well.

Chapter 4

Conclusions

4.1 Principal Conclusions

This thesis developed, validated, and deployed a laboratory-based Decision Support System for food contaminant risk surveillance, integrating SQL-based data engineering, statistical enrichment testing, machine learning classification, dual-evidence tiered scoring, bidirectional temporal validation, and One Health biological network analysis within a single automated, Docker-containerised, Papermill-orchestrated pipeline that is currently running in production at Neutron Laboratories, a Cotecna Group company.

The pipeline processed 182,033 certified analytical results across 122 contaminants and 640 food submatrices, amounting to 3,495 unique contaminant–submatrix pairs, covering the two-year window from January 2024 to December 2025 under Regulation (EU) 2023/915. Section-stratified Fisher exact testing with Benjamini–Hochberg correction identified 75 significantly enriched pairs out of the 3,496 pairs with at least five observations tested at the exploratory stage, one more than the 3,495 pairs that met the stricter modelling threshold and entered the DSS, with the most extreme effect sizes traceable to well-understood mechanisms rather than statistical artefacts: polychlorinated biphenyl biomagnification in fresh fish, at an odds ratio of 392.76, and cadmium bioaccumulation in fresh fish, at an odds ratio of 90.05. Section 2, Plant toxins, at an overall non-conformity rate of 16.1%, was the only section to cross the 5% alert threshold and stands as the most operationally urgent descriptive finding of this work, independently of any subsequent statistical or predictive modelling.

The CatBoost classifier, trained on eight leakage-free topological features, achieved a cross-validated PR-AUC of 0.8360, a held-out test PR-AUC of 0.8107, a test ROC-AUC of 0.9504, a 4.43-times lift over a dummy baseline, and a cross-validation-

to-test gap of 0.0254, comfortably below the pre-specified overfitting threshold, and it outperformed XGBoost, Random Forest, and LightGBM on every cross-validated comparison conducted. SHAP analysis identified two purpose-built topological features, `submat_nc_breadth` and `contam_nc_breadth`, as overwhelmingly dominant, together contributing more than four times the combined importance of the remaining six features, establishing firmly that the structural pattern of where non-conformity tends to cluster across the contamination landscape is a stronger predictor of any individual pair's risk than that pair's own specific identity or sampling history.

The dual-evidence scoring rule classified 127 pairs, 3.6% of the dataset, as HIGH risk, 248 pairs, 7.1%, as MEDIUM, 1,143 pairs, 32.7%, as LOW, and the remaining 1,977 pairs, 56.6%, as INSUFFICIENT for confident classification. The HIGH tier resolves into three largely distinct and mechanistically coherent clusters: multi-mycotoxin co-contamination centred on maize germ, pyrrolizidine alkaloid contamination across several botanical matrices, and lipophilic bioaccumulation of mercury and polychlorinated biphenyls in fresh fish. Most significantly, the frozen model and frozen DSS thresholds, never retrained and never recalibrated after their initial derivation on the 2024–2025 training window, were evaluated on two independent blind windows lying on opposite sides of that training period, prospectively on January–May 2026 and retrospectively on the whole of 2023; the evaluation subset for each window was built using support and non-conformity parameters derived separately and empirically from that window's own data, since a five-month and a twelve-month window warrant different treatment of how much evidence constitutes a stable signal, but the model itself and its two DSS tier thresholds were never adjusted in either direction, and both evaluations not only satisfied the pre-specified acceptance criteria but numerically exceeded the training reference values for both PR-AUC and ROC-AUC.

The One Health biological analysis, querying the 47 contaminants assigned HIGH or MEDIUM tier against the Comparative Toxicogenomics Database in both its human and mouse curated interaction sets, returned 405 gene targets, 261 of which qualify as hub genes shared by two or more contaminants, and 263 enriched KEGG pathways, of which 252, 95.8%, are shared across multiple contaminants. *ABCB1*, the gene encoding P-glycoprotein, together with the broader “ABC transporters” pathway, reached by 36 of the 47 contaminants, identifies a shared detoxification bottleneck whose simultaneous saturation by several co-occurring dietary contaminants constitutes exactly the kind of combinatorial toxicity mechanism that an assessment framework built around one contaminant and one maximum level at a time has no means of detecting.

4.2 Novelty and Scientific Contribution

Most MSc theses in bioinformatics centre on sequence data, structural prediction, or omics-scale molecular profiling. This one starts from regulatory compliance records generated in a production accredited laboratory, processes them through statistical enrichment testing and supervised machine learning, and only at its final stage reconnects the output to a molecular substrate through network biology. Statistical, predictive, and biological evidence converge on the same 47 contaminants, producing a risk prioritisation that is defensible at every level of the argument simultaneously; none of the three layers alone would carry the same evidential weight.

The contribution rests on three claims, each addressing a gap identified explicitly in the introductory chapter.

4.2.1 Translation of routine compliance data into prospective, production-deployed risk intelligence

LIMS data is generated, in the overwhelming majority of accredited laboratories, purely as a by-product of pass-fail compliance decisions and is rarely analysed in aggregate once reported. This thesis turns 182,033 such records into a continuously updated, tier-stratified risk ranking spanning 3,495 contaminant–submatrix combinations, through a pipeline that is not a one-off research exercise but a Docker-containerised, Papermill-orchestrated system running in production at Neutron Laboratories today, reading directly from the live LIMS described in Chapter 2. No external public resource, not RASFF, not EFSA’s occurrence data, not any national monitoring programme, matches the granularity, recency, and market-specific precision of this proprietary occurrence data; the contribution is a working mechanism for actually exploiting it.

4.2.2 A dual-evidence scoring methodology validated bidirectionally in time

The DSS tiering rule requires agreement between two independently derived evidence sources, a frequentist enrichment test and a supervised machine learning probability, directly operationalising the asymmetric cost of the two error types that matter most in food safety monitoring. Its eight-feature set is leakage-free by construction, with the two dominant features topological rather than pair-specific, drawing on the structure of the entire contamination landscape rather than any single pair’s own narrow history. The validation is unusually demanding: rather than a single

forward-looking prospective test, the frozen model and thresholds were evaluated in both temporal directions at once, on a window following training and one preceding it, and both exceeded the training reference. This bidirectional agreement is difficult to attribute to drift, recency, or temporal good fortune; a transient artefact of the 2024–2025 training window specifically would not be expected to survive being tested against 2023 in this way.

4.2.3 A One Health biological layer that grounds the DSS output in molecular mechanism

Combinatorial toxicity through shared molecular targets is a structural blind spot of any regulatory framework that assesses contaminants strictly one at a time. For the specific 47-contaminant portfolio that the DSS identifies as HIGH or MEDIUM risk, this thesis constructs a contaminant-gene-pathway network grounded in curated CTD evidence, naming the hub genes, with *A2M*, *ABCB1* and *AARS1* foremost among them, and the shared pathways, “Lipid and atherosclerosis” and “ABC transporters” foremost, through which this portfolio converges biologically rather than acting as forty-seven independent risks. This is not offered as proof of a measured physiological interaction, but as a biologically motivated rationale for the DSS’s prioritisation, and it identifies *ABCB1*-mediated efflux competition specifically as the single most actionable mechanistic finding from this analysis, pointing directly towards the scoring extension discussed in Section 4.4.

Together, these three contributions trace a path from a raw, individually-assessed compliance record through to a biologically contextualised, production-deployed risk intelligence system.

4.3 Limitations

Several limitations of this work should be stated explicitly rather than left implicit. The pipeline, and in particular the two fixed DSS thresholds, $DSS_HIGH = 0.062$ and $DSS_MED = 0.009$ (the 90th and 75th percentiles of the observed non-conformity-rate distribution rather than absolute rates fixed a priori), is calibrated specifically to the contaminant–submatrix portfolio and analytical volume of Neutron Laboratories, and there is no guarantee that these same threshold values would transfer directly to a different laboratory with a different client portfolio without being re-derived on that laboratory’s own training distribution. The labelling rule underlying the machine learning target is sensitive to the length of the observation window used, as demonstrated directly by the decline in positive rate from 18.8% in the training set to

7.8% over twelve months to 2.9% over five months; while this relationship was shown to be an internally consistent artefact of the labelling design rather than evidence of model failure, it does mean that raw HIGH-tier counts on a short evaluation window should be interpreted with this artefact explicitly in mind, even though the underlying PR-AUC and ROC-AUC ranking metrics themselves remain valid regardless of window length. The calibration analysis identified mild overconfidence specifically in the 0.4 to 0.6 predicted probability range, so the model should be treated as a reliable ranking instrument throughout its output range, but as a trustworthy absolute probability estimate primarily above roughly 0.6. The One Health biological layer is, at present, purely contextual and informational; hub gene degree and pathway convergence do not currently feed back into, or modify in any way, the DSS composite score itself. Finally, and perhaps most importantly given how the biological results are framed throughout Chapter 3, the underlying CTD interaction data pools curated evidence from both human and mouse experimental models into a single combined gene set per contaminant, without distinguishing in the resulting network which specific interactions were established in which organism, and generally without recording the specific dose or route of administration of the original experimental evidence; the data therefore does not by itself constitute evidence of an interaction occurring at realistic dietary exposure levels in humans specifically. Translating from a CTD-documented mechanism, established in either organism, to a population-level human dietary risk claim would require epidemiological corroboration that lies outside the scope of this thesis, and the biological findings reported here should be read as establishing mechanistic plausibility and convergence, not as establishing a demonstrated physiological effect at dietary doses in humans. A related limitation applies to the seasonal pattern discussed in Section 3.2: the LIMS records a sample's arrival date, not its harvest date, so the coincidence between the Section 1 non-conformity peak and the maize harvest window cannot be verified at the level of individual samples.

4.4 Future Directions

Several extensions follow naturally and directly from the results presented in the previous sections of this work. The most immediately actionable, given the magnitude of the *ABCB1* and broader ABC transporter signal identified in Section 3.8, would be to incorporate a measure of hub-gene degree or pathway-level convergence as an additional weighted term in the composite DSS score, alongside the existing non-conformity rate and model probability terms, producing a scoring system that accounts explicitly for combinatorial toxicity potential rather than treating the biological layer as a separate, downstream commentary on a score already fixed.

The pipeline's configuration already anticipates a scheduled retraining once a full cumulative 2024–2026 dataset becomes available, which would extend the training window, allow the DSS thresholds to be recomputed on a correspondingly larger base, and provide a further, independent test of whether the bidirectional validation result reported in this thesis continues to hold once the training distribution itself has grown. Incorporating product geographic origin as an additional model feature would capture the well-documented geographic determinism of heavy metal soil contamination and of climate-driven mycotoxin risk, and represents the single highest-value feature addition available directly from existing LIMS fields, requiring no new data collection effort. Extending validation to other laboratories within the Cotecna Group would simultaneously test the external validity of the fixed DSS thresholds across a genuinely different client portfolio and substantially enlarge the training base available for the long tail of contaminant–submatrix combinations that are currently, and will likely remain for some time, classified as INSUFFICIENT. Finally, extending the underlying pipeline to cover pesticide residues under Regulation (EC) No 396/2005, by far the numerically dominant analytical workload at most accredited food testing laboratories, would require only an extension of the regulatory limit mapping component described in Section 2.2; every downstream statistical, machine learning, and biological module described in this thesis is directly reusable without structural modification.

Appendix A

Supplementary Information

This appendix is restricted to material that directly extends results already reported in Chapter 3: the raw confusion counts behind the summary metrics in Table 3.6, the full sensitivity grid behind the single value quoted in Section 3.4.1, the precision-recall and ROC curves behind the point estimates in Section 3.4.2, the pair-level diagnostic supporting the false-positive discussion in Section 3.4.3, the confusion matrix at an alternative decision threshold not discussed in the main text, and the pathway clustering pattern underlying the shared-pathway analysis in Section 3.8.

A.1 Supplementary Table

Table S1: Raw confusion counts for both blind validation windows, complementing the summary metrics given in Table 3.6. Sources: `outputs_2026/NB_blind_dss_scores_2026.csv` and `outputs_2023/NB_blind_dss_scores_2023.csv`.

	2026 prospective ($n = 751$)	2023 retrospective ($n = 668$)
True positives	21	42
False negatives	1	10
False positives	21	14
True negatives	708	602

A.2 Supplementary Figures

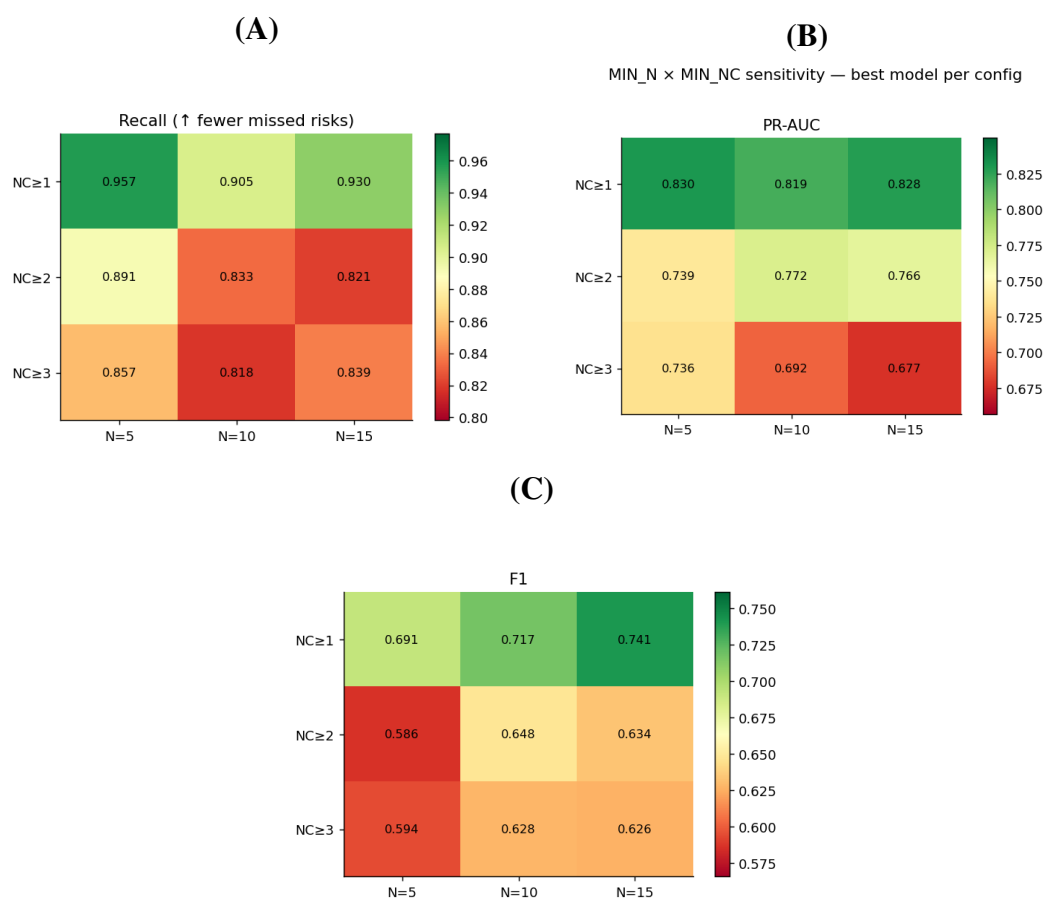


Figure S1: Performance of the best-performing model at each of the nine MIN_N $\times$ MIN_NC configurations: (A) recall; (B) PR-AUC; and (C) F_1 . The grid complements the single value quoted for the selected configuration in Section 3.4.1.

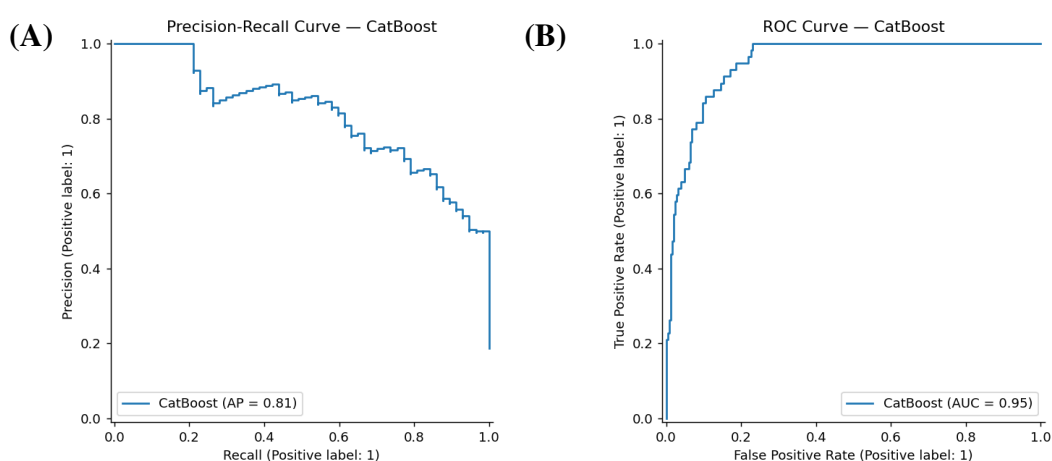


Figure S2: Performance curves for the final CatBoost model on the held-out test set: (A) precision–recall curve (AP = 0.81); and (B) ROC curve (AUC = 0.95). These curves underlie the point estimates reported in Section 3.4.2. Precision remains above 0.85 for recall values up to approximately 0.4, reflecting high confidence on structurally well-supported pairs.

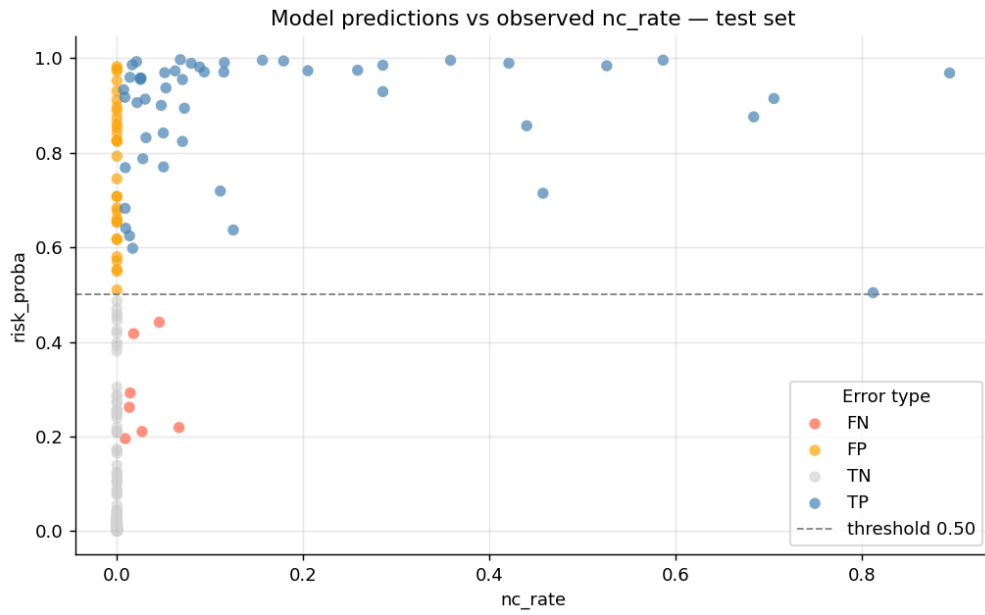


Figure S3: Predicted risk probability against observed non-conformity rate for all 304 pairs in the held-out test set, coloured by prediction outcome (TN, FP, FN, TP), complementing the tier-coloured view of the same two quantities in Panel A of Figure 3.5. False positives are concentrated at low non-conformity rate with high predicted probability, supporting the discussion in Section 3.4.3: these pairs score high on structural breadth features despite having no direct non-conformity history within the observation window.

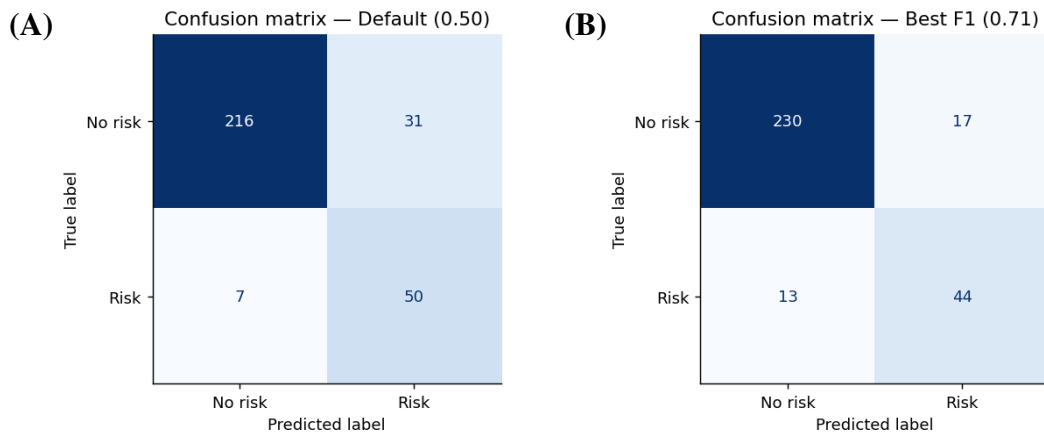


Figure S4: Confusion matrices on the held-out test set at two decision thresholds. (A) Default threshold 0.50 (TN = 216, FP = 31, FN = 7, TP = 50), as used throughout the DSS scoring rule. (B) Best- F_1 threshold 0.71 (TN = 230, FP = 17, FN = 13, TP = 44), not otherwise discussed in the main text, showing the precision–recall trade-off available if the classification threshold were moved.

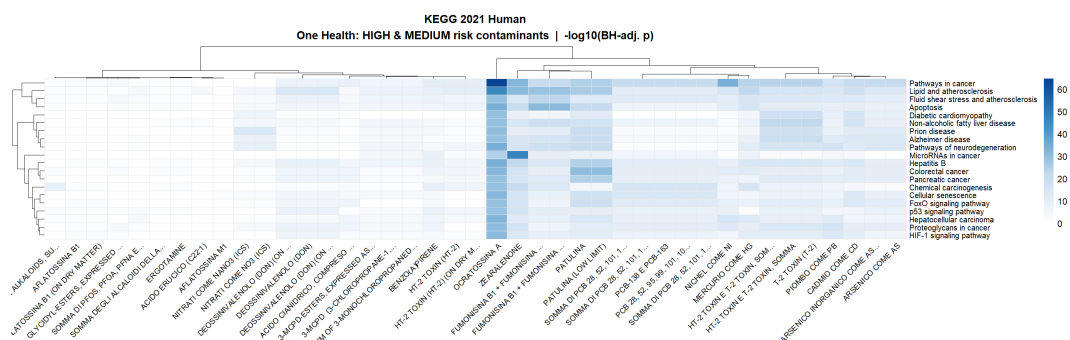


Figure S5: Hierarchically clustered heatmap of KEGG 2021 Human pathway enrichment ($-\log_{10}$ BH-adjusted p) for the 47 HIGH and MEDIUM tier contaminants, underlying the shared-pathway discussion in Section 3.8. The two main contaminant clusters correspond broadly to mycotoxins and heavy metals, yet both share the dominant pathway block comprising lipid and atherosclerosis, apoptosis, and pathways in cancer.

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